

Swing 3

User Manual · Version 3.0

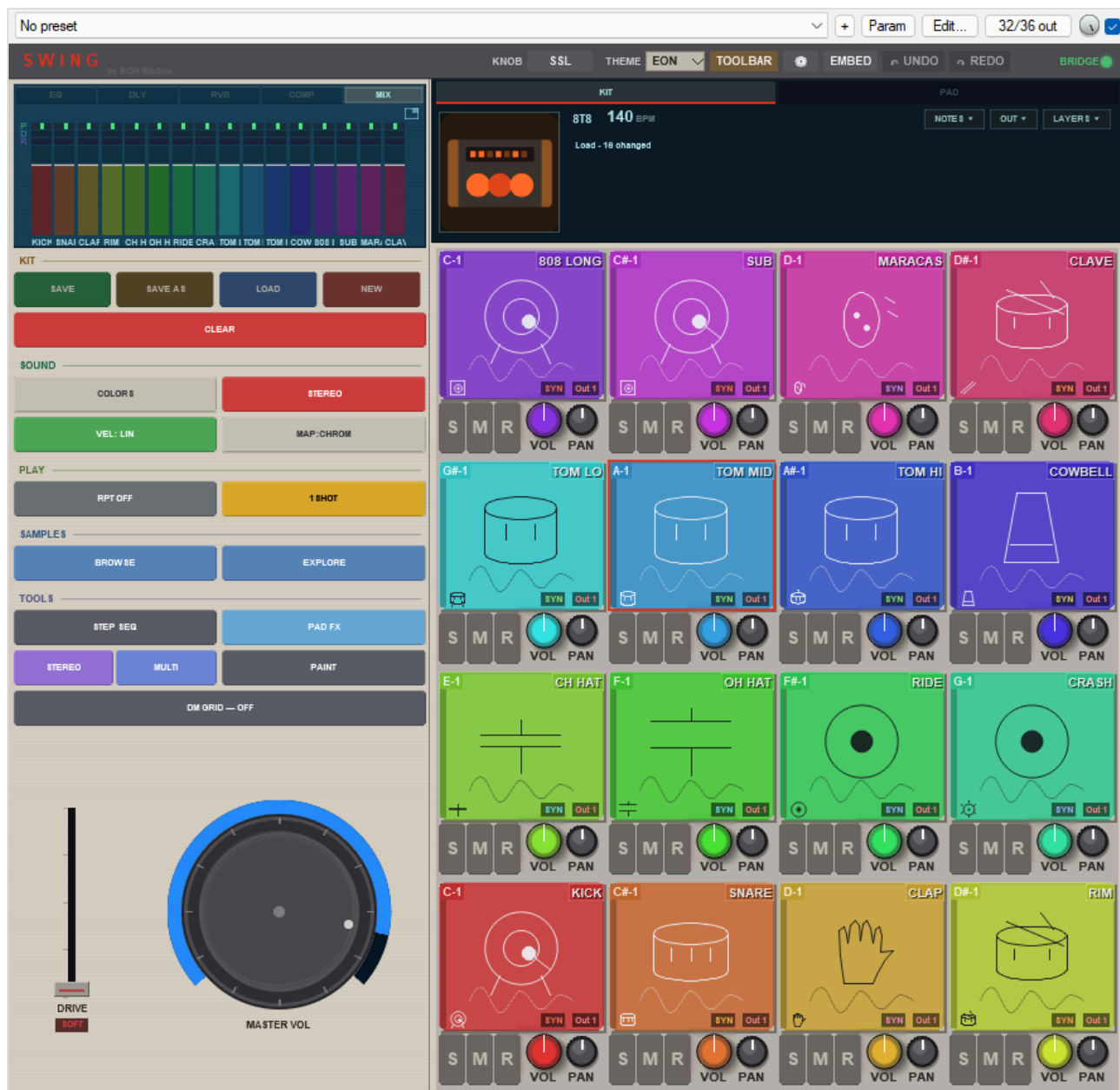
EON Studios

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1. Overview

Swing is a 16-pad drum sampler instrument built as a JSFX plugin for REAPER. Each pad can play a sample (or up to four layers), shaped by an ADSR envelope and routed to any of 16 stereo output buses. Every pad has its own effects chain — filters, EQ, saturation, bitcrush and delay/reverb sends — and the left-side LCD tabs expose master EQ, delay, reverb, compressor and mixer controls.



What's new in v3. Three changes are big enough to alter how you work:

- **Every pad can be a synth.** Nine engine families, thirty voice types, no samples required. See chapter 6.
- **The pad grid doubles as a workspace.** Ten full-window views open over it — note mapping, colours, routing, the kit library, the synth editor, the mixer and the four master FX pages. See chapters 9 and 10.
- **Every instance is its own instrument.** Run as many Swings as you like with no cross-talk. See chapter 16.

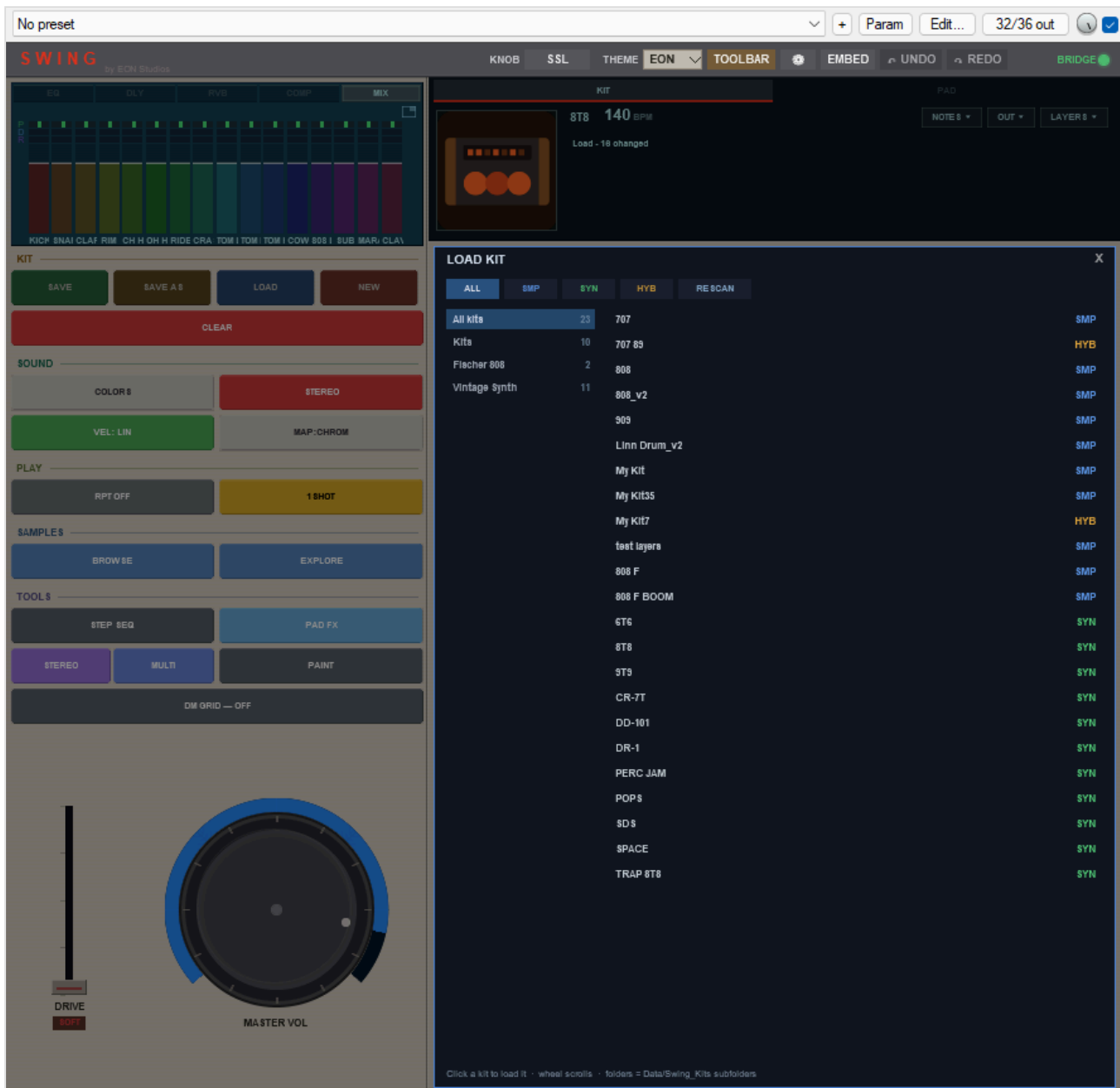
Swing integrates with two background helpers — the **Swing Kit Bridge** (a Lua service handling kit save/load, batch import, chop-to-pads and relinking) and the **Swing Browser** (an ImGui sample browser). Version 3 adds three companion plugins: the **Step Sequencer**, the **Drum Matrix** and the **Drum Strip**.

Requirements. REAPER 7.0 or later, plus three ReaPack extensions: **ReaImGui** (the sample browser), **SWS** (a few sync actions) and **js_ReaScriptAPI** (the Drum Matrix). Swing's JSFX itself needs none of them.

2. Quickstart

Load Swing on a track. Insert `Swing_ReaKit` from the FX browser (`EON` → `Swing`). It opens on the last kit you used, or a factory kit on a fresh install.

Load a kit. Click **LOAD**. This opens the Kits view, where the factory library and your own kits are browsable by folder.



Load your own samples. Drag audio files onto pads, from the Swing Browser or REAPER's Media Explorer. Multi-file drags fill consecutive pads. Pads auto-name and auto-categorise by drum type where the filename allows it.

Play from MIDI. Pads map to a root note (default C1 = 36) plus pad index. Record or draw MIDI notes in the MIDI editor, or click a pad to audition it.

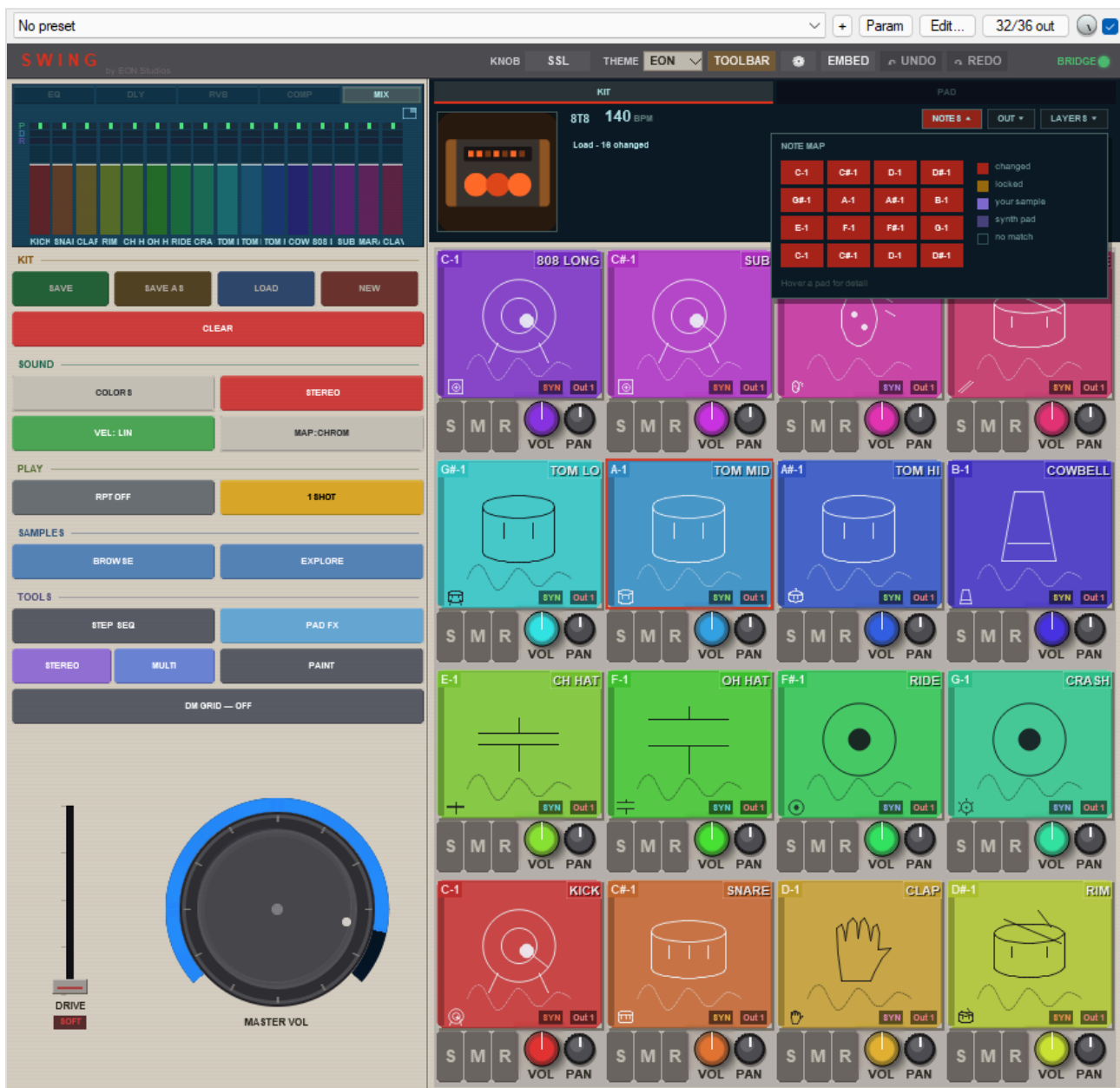
Save your kit. **SAVE** overwrites; **SAVE AS** prompts for a name. A `.swing` file stores pad assignments, tuning, levels, colours, routing, envelopes, FX settings, synth voices and the cover image.

3. User Interface Tour

Pad grid. Sixteen pads show colour, name, category glyph, note, output bus and a waveform thumbnail. Below each pad is a control strip: **S** solo, **M** mute, **R** repeat, and VOL and PAN knobs. Click a pad to trigger it; right-click for pad actions.

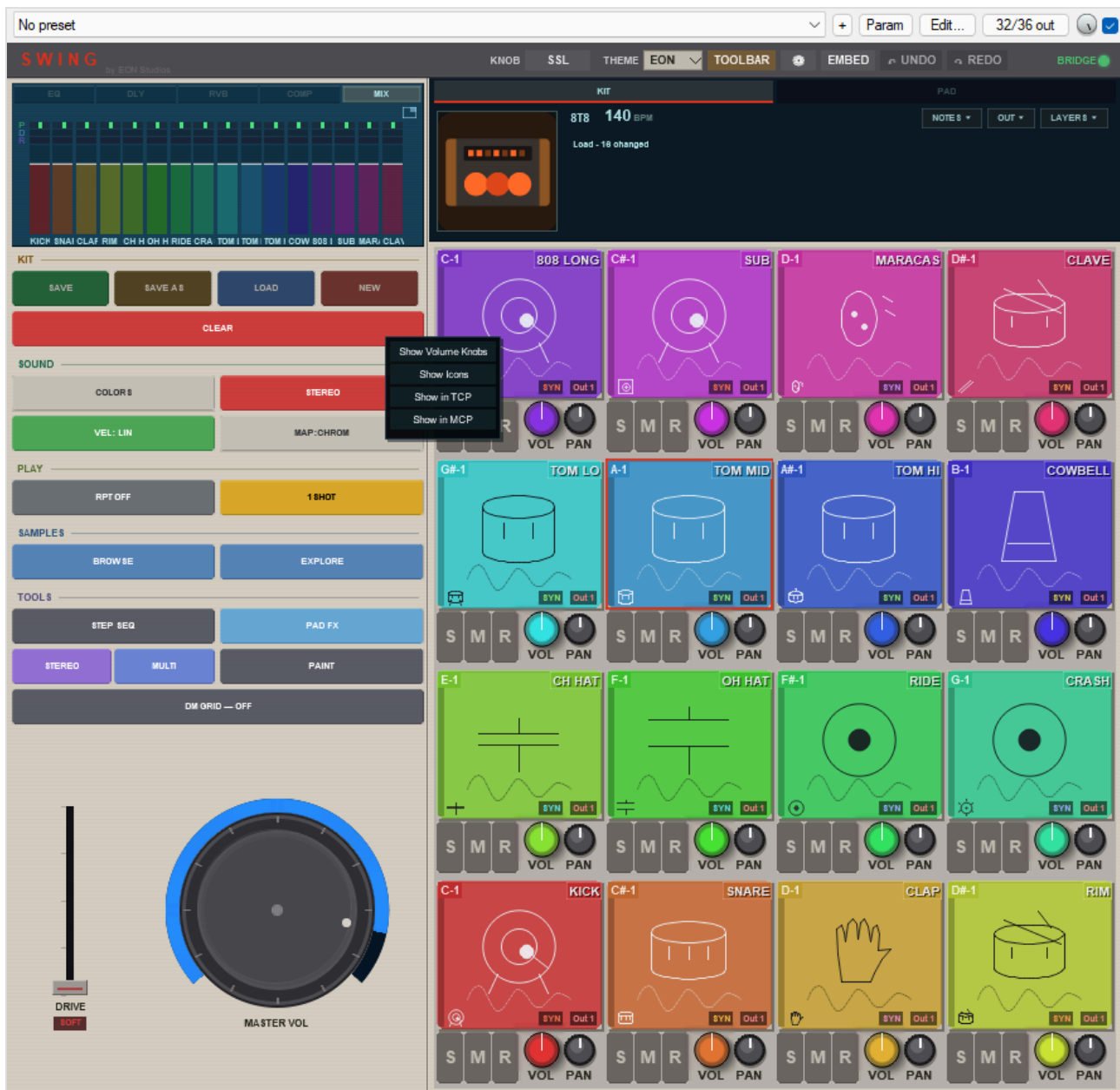
Left panel. Kit controls (SAVE, SAVE AS, LOAD, NEW, CLEAR), sound options, the tools row, and master volume and drive.

Right panel. Switches between **KIT** and **PAD** tabs. The KIT tab shows the cover image, project tempo and three dropdown panels — NOTES, OUT and LAYERS.



LCD tabs. Along the top left — EQ, DLY, RVB, COMP, MIX. These are the master processing pages and apply globally. See chapter 10.

Embedded mode. Shown in REAPER's track panel (TCP) or mixer (MCP), Swing renders a compact pad grid for quick access. The gear menu chooses what it shows and where — volume knobs, category icons, TCP, MCP:



The leftmost channel in the mixer figure in chapter 14 shows the result — Swing's compact grid living in the MCP beside its own drum channels.

4. Pads & Playback

Gain / Pan / Tune. Per-pad knobs set level, stereo position and pitch in semitones. In Repitch mode the tune knob responds immediately. Keep an eye on master output to avoid clipping.

ADSR. Attack shapes the transient, decay and sustain shape the body, release controls the tail after note-off. For one-shots, keep release moderate to avoid clicks.

Offset. Moves the playback start point into the sample. This is the single most useful control for tightening a kick: push past the silence at the front and the hit lands where you put it.

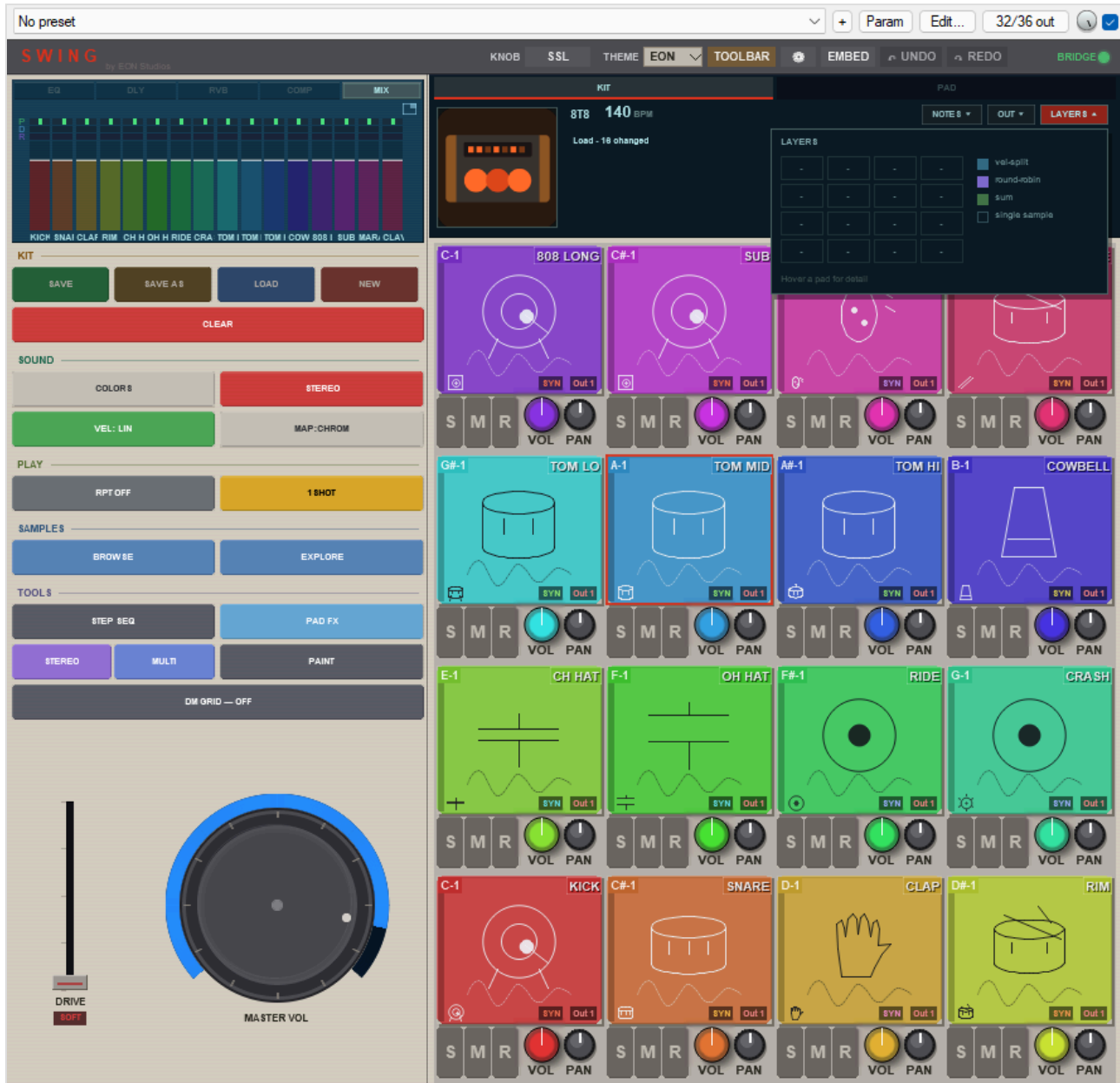
Reverse. Plays the pad backwards, for transitions and risers.

One-shot. Ignores note-off and plays to completion — right for drum hits, wrong for anything you want to cut short.

Choke groups. Assign pads to a group (1–8) so triggering one stops another. The classic use is an open and closed hi-hat sharing a group.

5. Layers

Each pad supports up to 4 layers. Layers can be used for velocity dynamics, round-robin variation, or stacked sounds.



Velocity splits. Set each layer's velocity low/high thresholds so different samples trigger at different hit strengths.

Round-robin. Cycle through layers on consecutive hits to reduce the machine-gun effect on fast rolls.

Sum mode. Play multiple layers together. Use tightness to control whether layers align perfectly or smear slightly for thickness — a little smear is how you widen a clap.

Per-layer trim and gain. Each layer has its own start/end trim and gain compensation, so you can level-match sources that were never recorded together.

6. The Drum Synth

New in v3.

Every pad carries a synthesised voice alongside its sample layers, or instead of them. **Nine engine families, thirty voice types:** kick, snare, clap, hat, ride, tom, rim, cowbell and shaker.



Pick a pad on the left, shape it on the right. The macro controls change with the voice — controls that do nothing for a given engine are hidden rather than greyed out, so what you see is what that voice responds to.

Above the controls sits a 3D rendering of the drum you are building. It is not decoration: it follows the macros and responds to velocity, so tuning the shell changes its shape and hitting harder moves it.

These are ports of hardware-measured references rather than approximations, so the classic voices behave the way the machines they came from behaved.

A synth pad is a full pad. It plays from MIDI and the step sequencer, takes the per-pad EQ, FX, choke groups and sends, saves into kits, and undoes like anything else.

The eleven Vintage Synth kits are built entirely on this engine — 8T8, 9T9, 6T6, CR-7T, SDS, DR-1, POPS, SPACE, TRAP 8T8, DD-101 and PERC JAM. No samples, so they load instantly, weigh almost nothing, and every voice stays editable.

7. Per-Pad FX

Swing gives every pad its own FX chain, edited on the PAD FX panel.



HPF / LPF. High-pass and low-pass filters (20 Hz–20 kHz) for cleanup and tone shaping.

EQ. With an interactive curve — drag the nodes. Start with subtle $\pm 1\text{--}3$ dB moves.

Saturation. Five drive modes: Soft, Tube, Tape, Fuzz, Clip. Use small amounts first; Clip is the most aggressive.

Bitcrush. Rate reduction plus bit depth, for lo-fi textures.

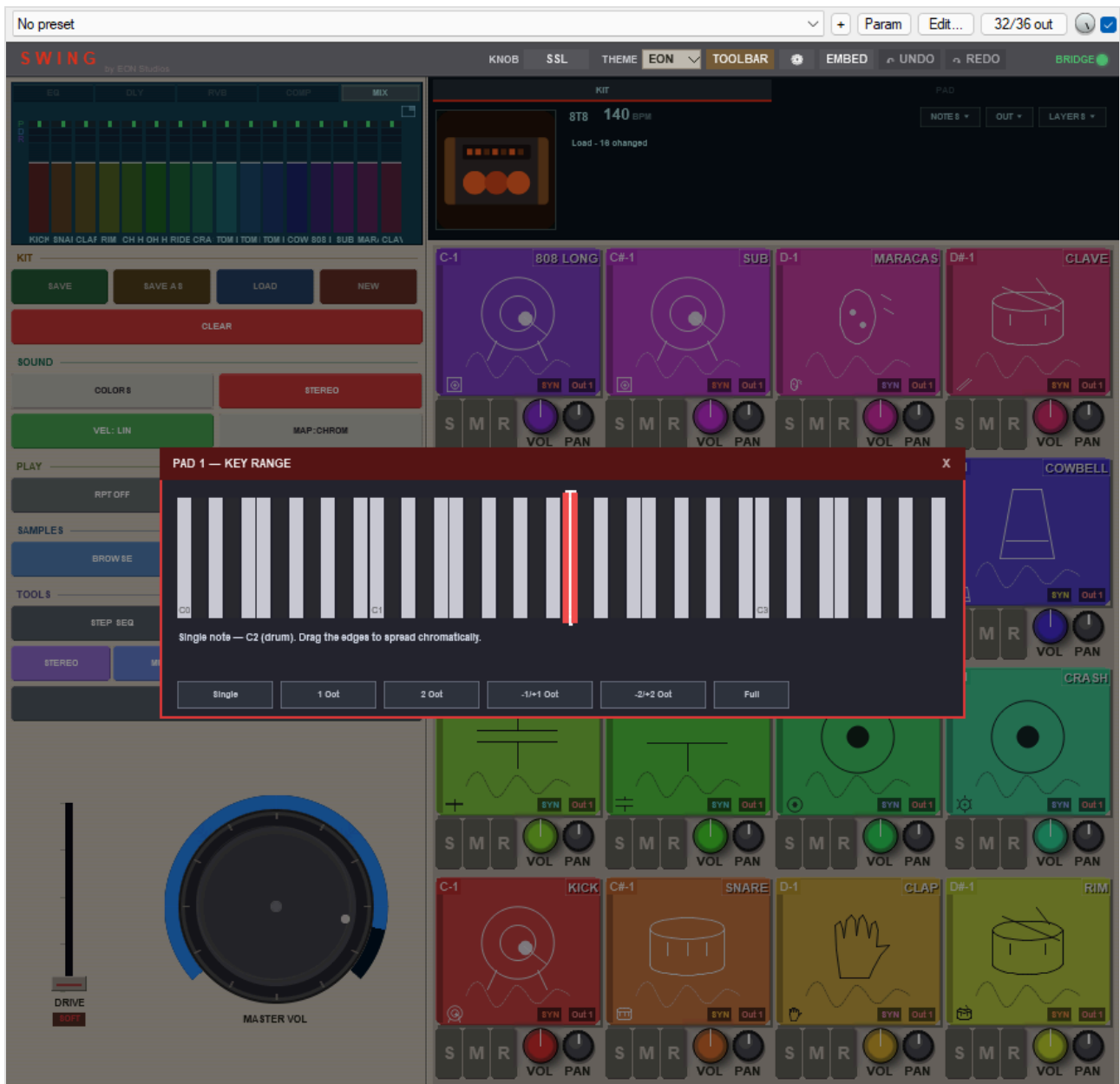
Delay / reverb sends. Send each pad into the global engines for cohesion.

8. Key Ranges & Tune Modes

New in v3.

Key ranges

Give a pad a key range and it plays across the keyboard instead of on one note.



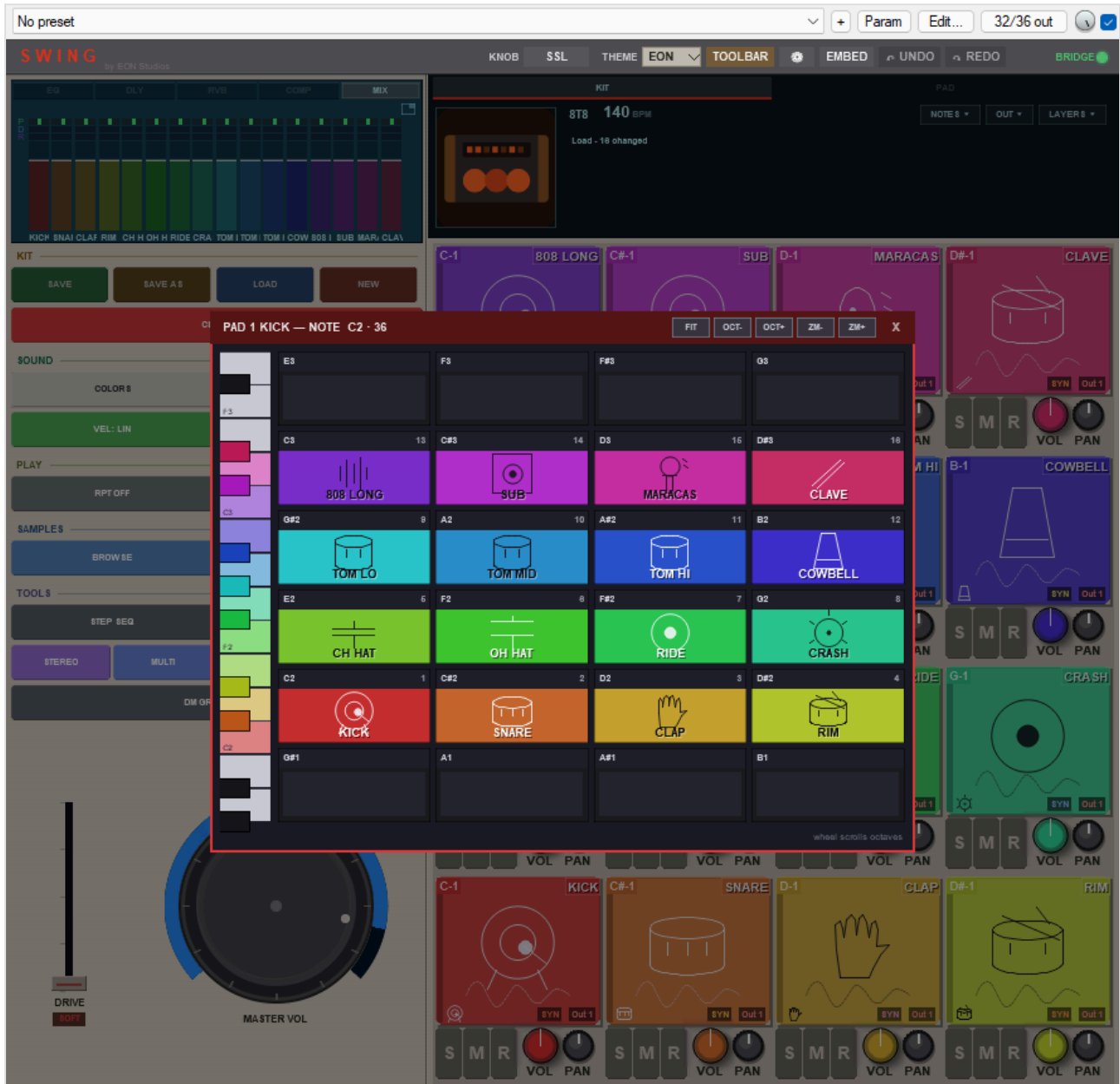
Right-click the **RANGE** pill to set how far the range reaches either side of the pad's own note. The range is stored as offsets from that note rather than as fixed keys, so it survives a kit change — load a new kit and your playable pad is still playable.

Tempo-repeat follows the key you are actually holding, so held-note rolls track the melody rather than hammering the pad's root.

Two limits worth knowing. A ranged pad is **monophonic** — it plays melodically across the keys, one note at a time, and does not play chords. And key ranges track in **Repitch** mode only; Tuned and Stretch support is still to come, and Swing says so in the range editor rather than leaving you to find out.

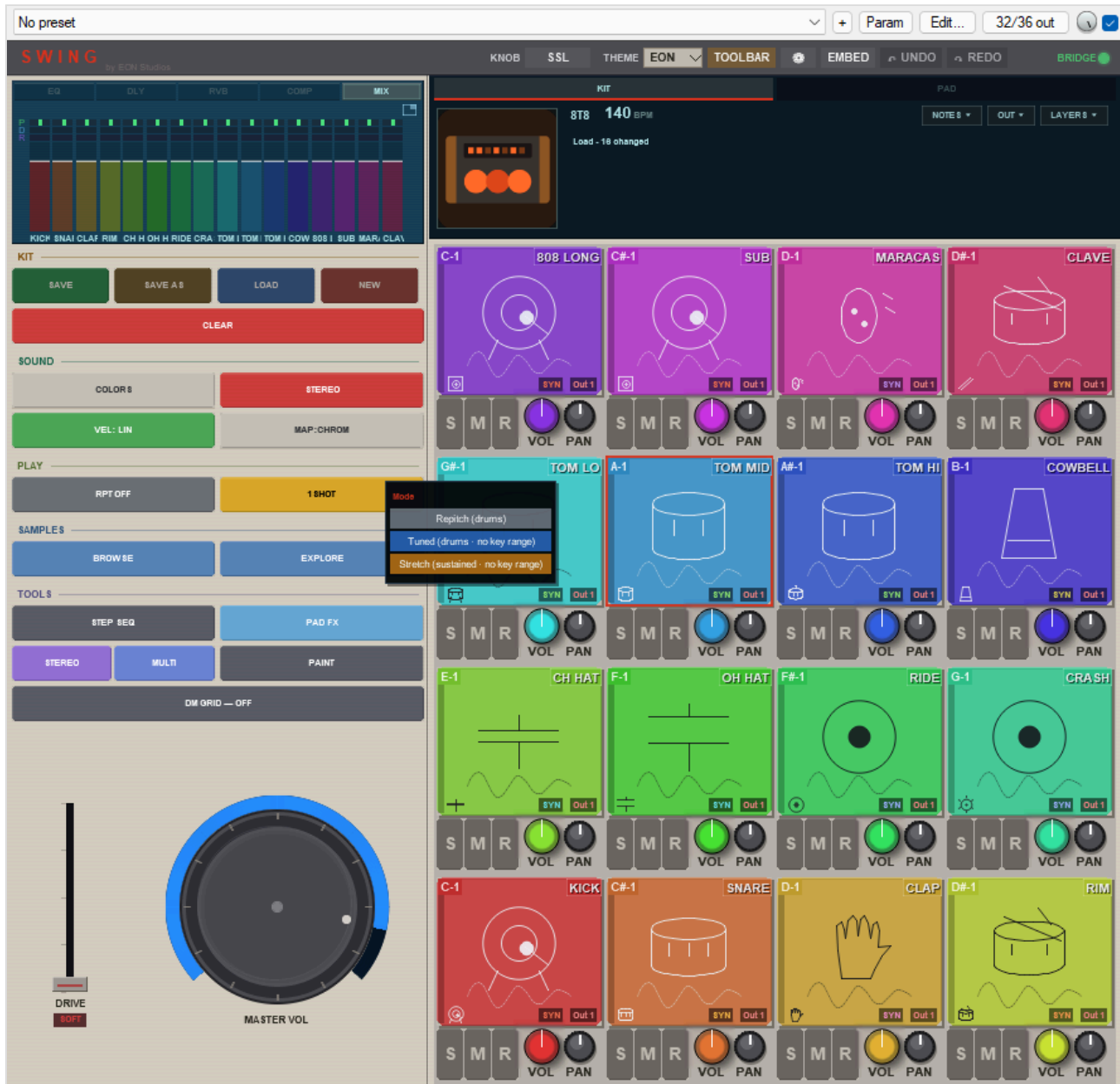
The note picker

Click the **NOTE** pill on a pad to open the note map framed on that pad.



Click a key to set the pad's note, or drag the pad's chip along the keyboard. It opens framed on your kit with an empty row above and below, so you can see where the pad sits before you move it.

Tune modes



Repitch is varispeed — pitch and length move together, like tape. Free, and right for drums.

Tuned is pitch-preserved, using a bake with per-layer anchors so velocity-layered pads pitch every layer correctly.

Stretch is real-time time-stretch, and requires the EON engine.

The hints in the picker steer sustained material toward Stretch and drums away from it. **Without the EON engine installed**, Tuned and Stretch play as Repitch, and Swing shows a notice rather than quietly sounding wrong.

9. The Big View

New in v3.

Ten views open over the pad grid, all bindable to keys and toolbar buttons, plus an eleventh action that closes whichever is open.

Mapping

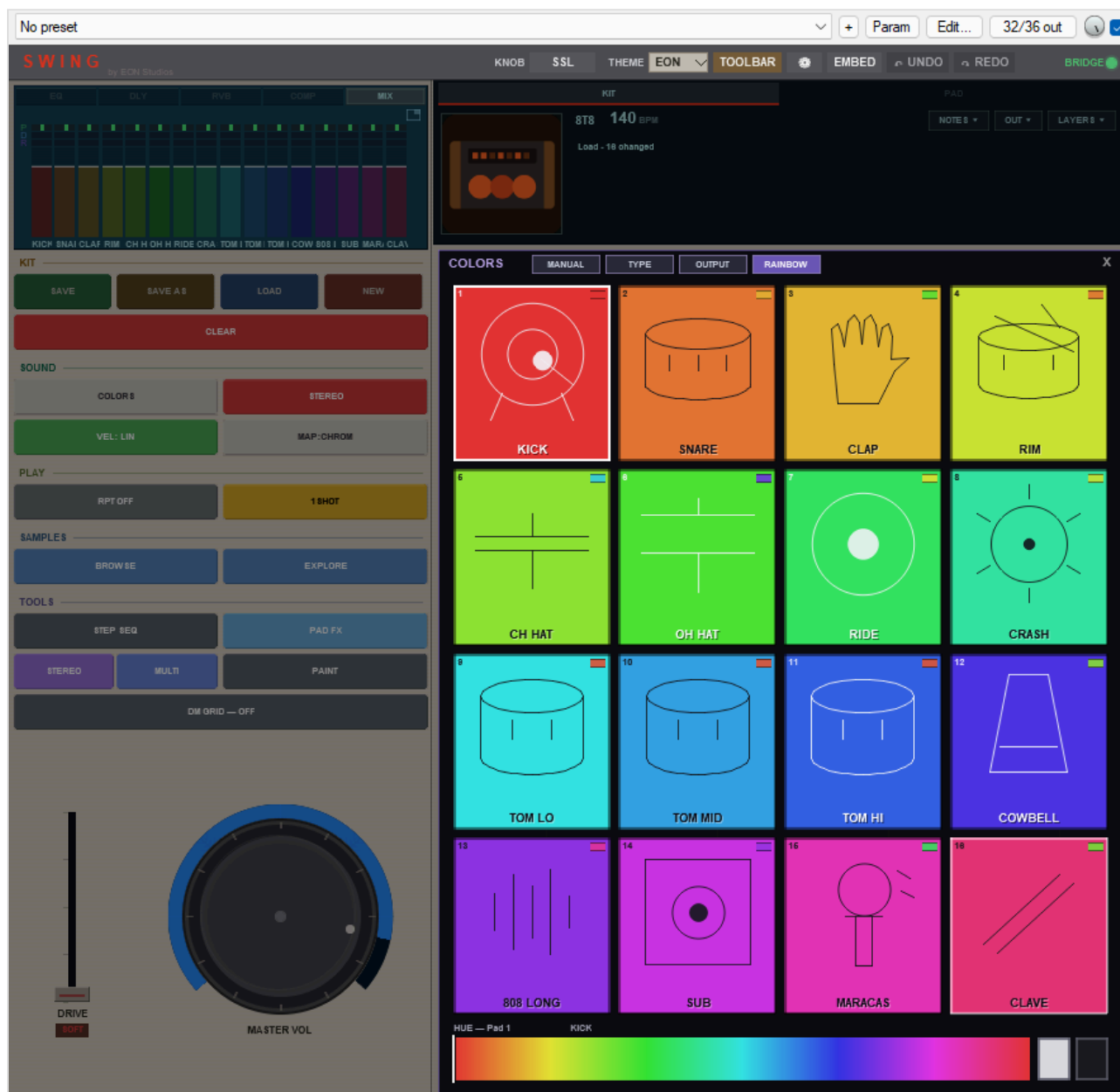


A piano-roll note map with draggable keys. Seven layouts cover the usual controller shapes:



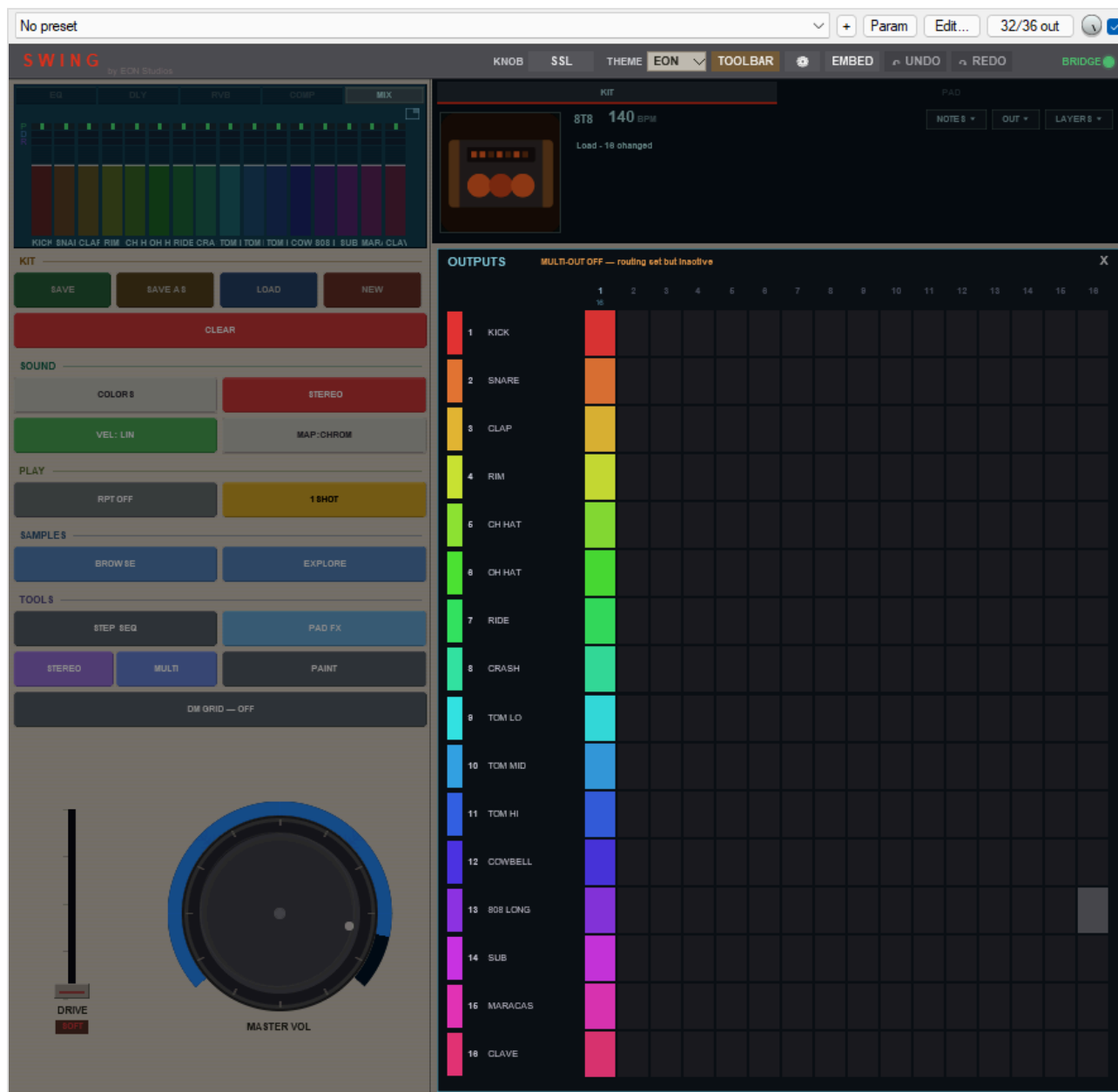
Bank A/B switches the lower and upper halves. The **choke lane** underneath is editable — that is where open and closed hats get grouped.

Colours



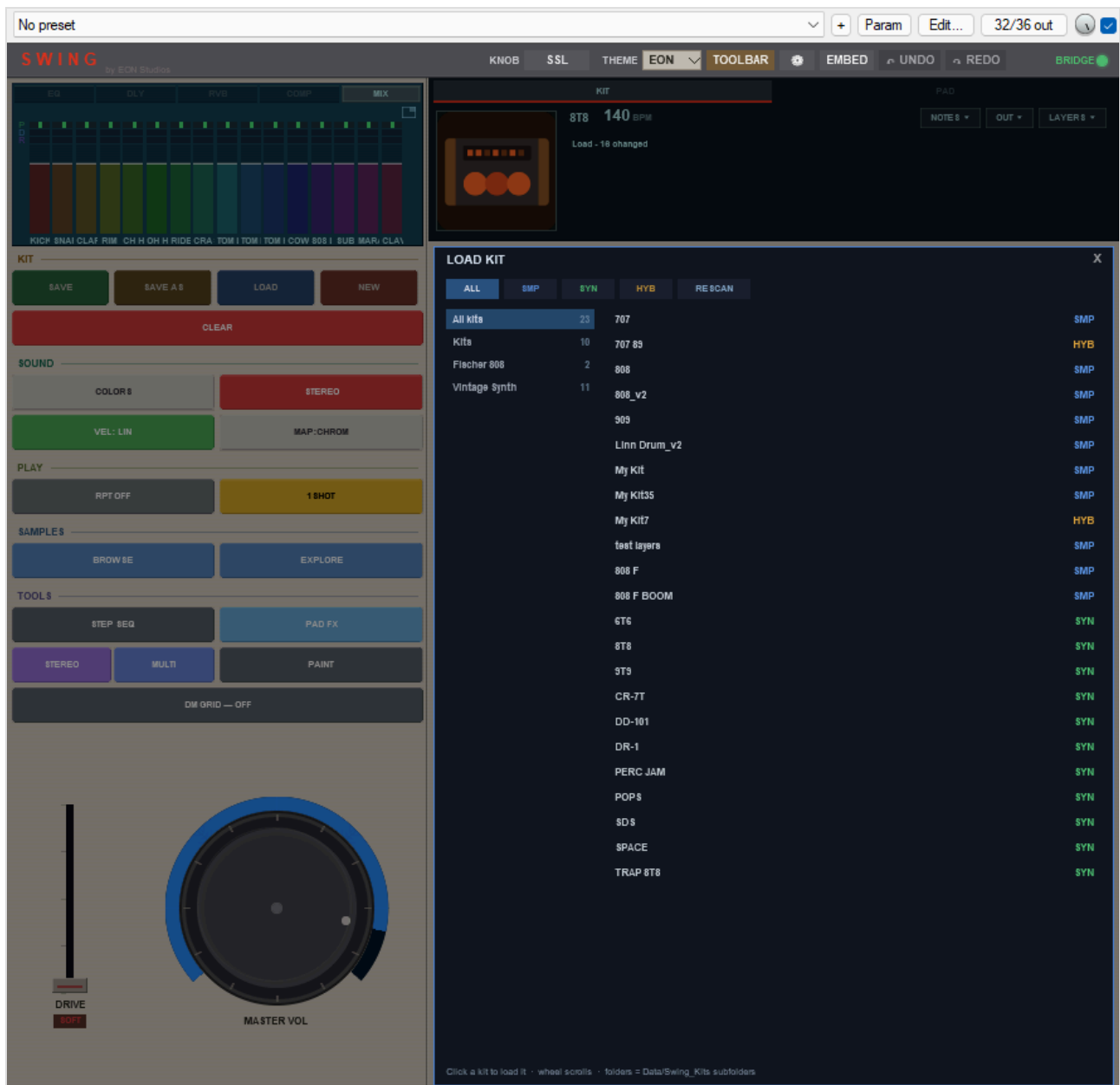
Four modes — MANUAL, TYPE, OUTPUT and RAINBOW — plus a hue strip for the selected pad. Pad colour carries through to track colours and sequencer lanes, so a kit stays recognisable everywhere it appears.

Outputs



The routing matrix. See chapter 14 for building a multi-out rig.

Kits



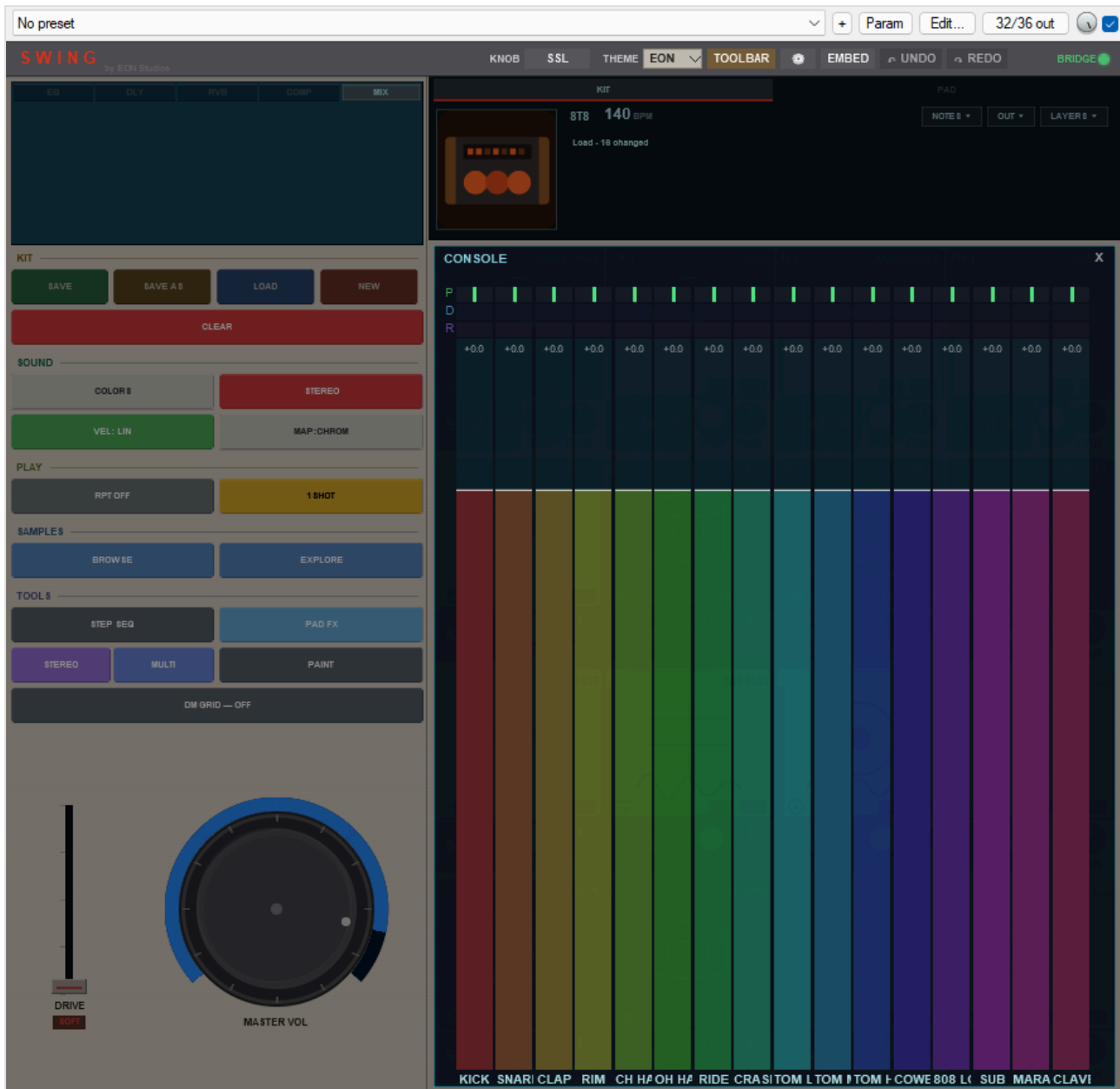
The library, browsable by folder and filterable by sample, synth or hybrid. Left-clicking **LOAD** comes here.

Coming from 2.1? The old *Load Kit (.swing)* entry is gone from the LOAD

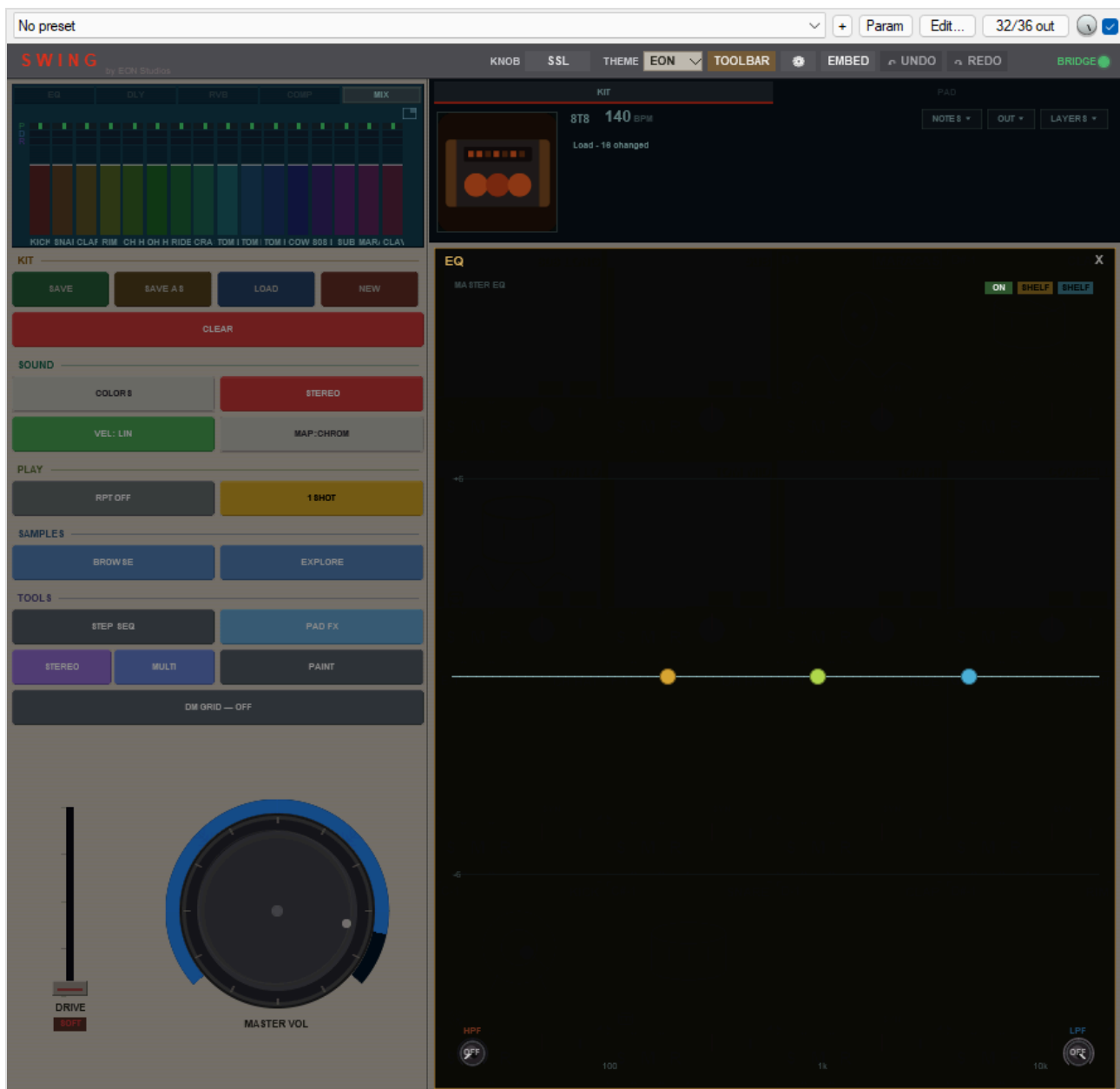
right-click menu. Loading a kit file lives in the Kits view now.

10. Global FX & Mixer

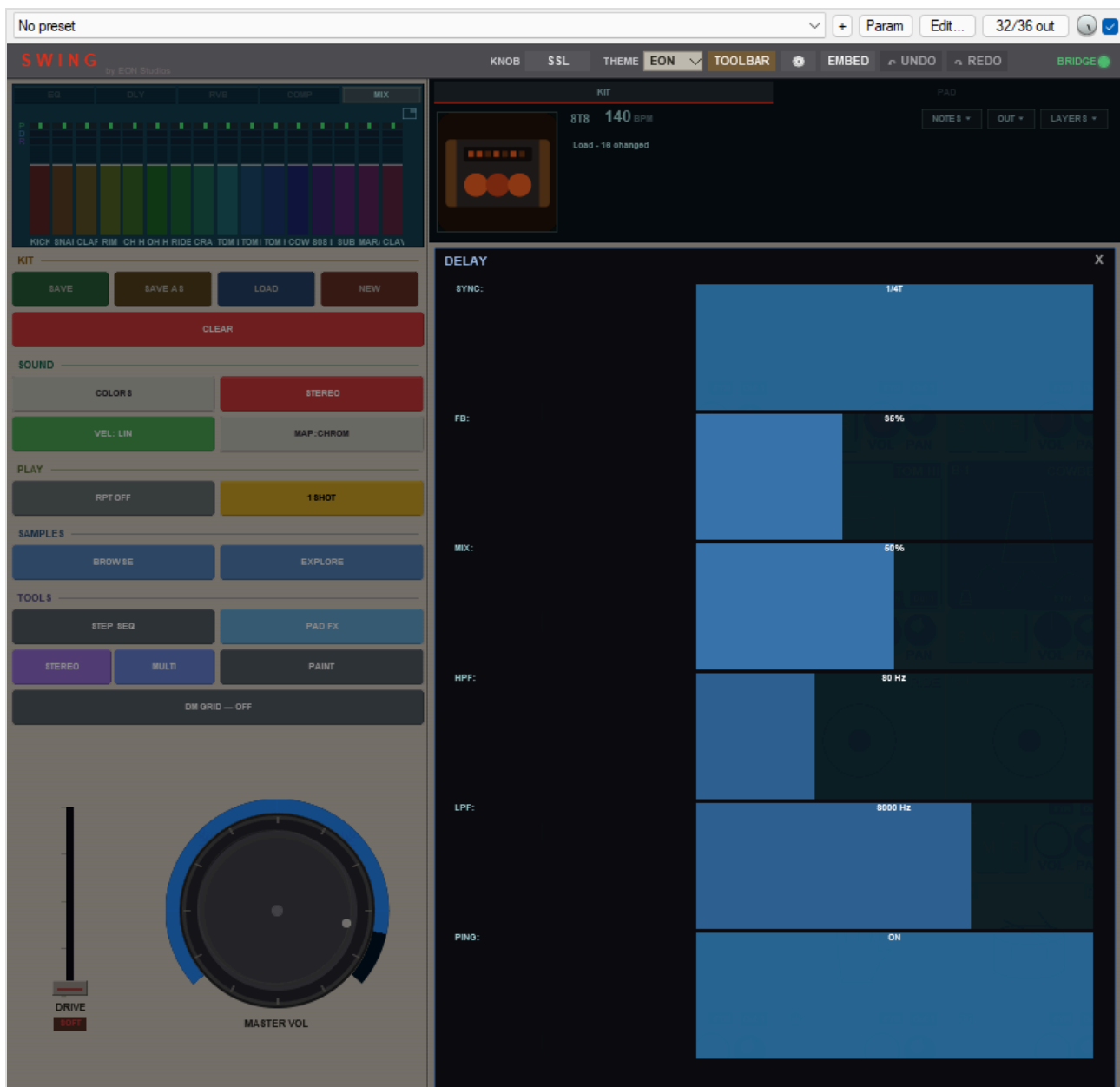
Use the LCD tabs for master processing.



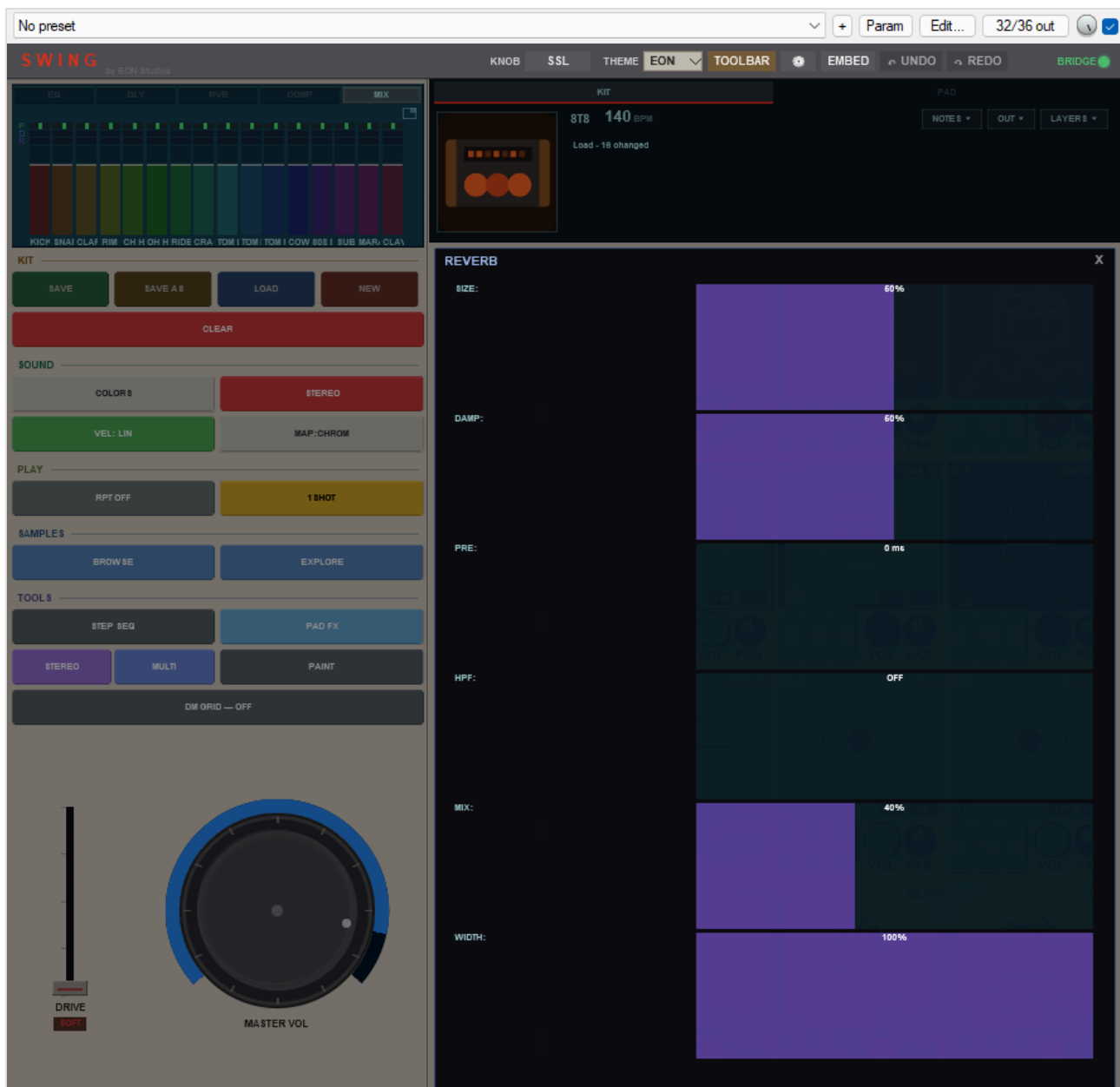
EQ. Four bands of zero-zipper state-variable filtering, with a live curve. Rebuilt in v3 — coefficients glide instead of stepping, so sweeping is silent.



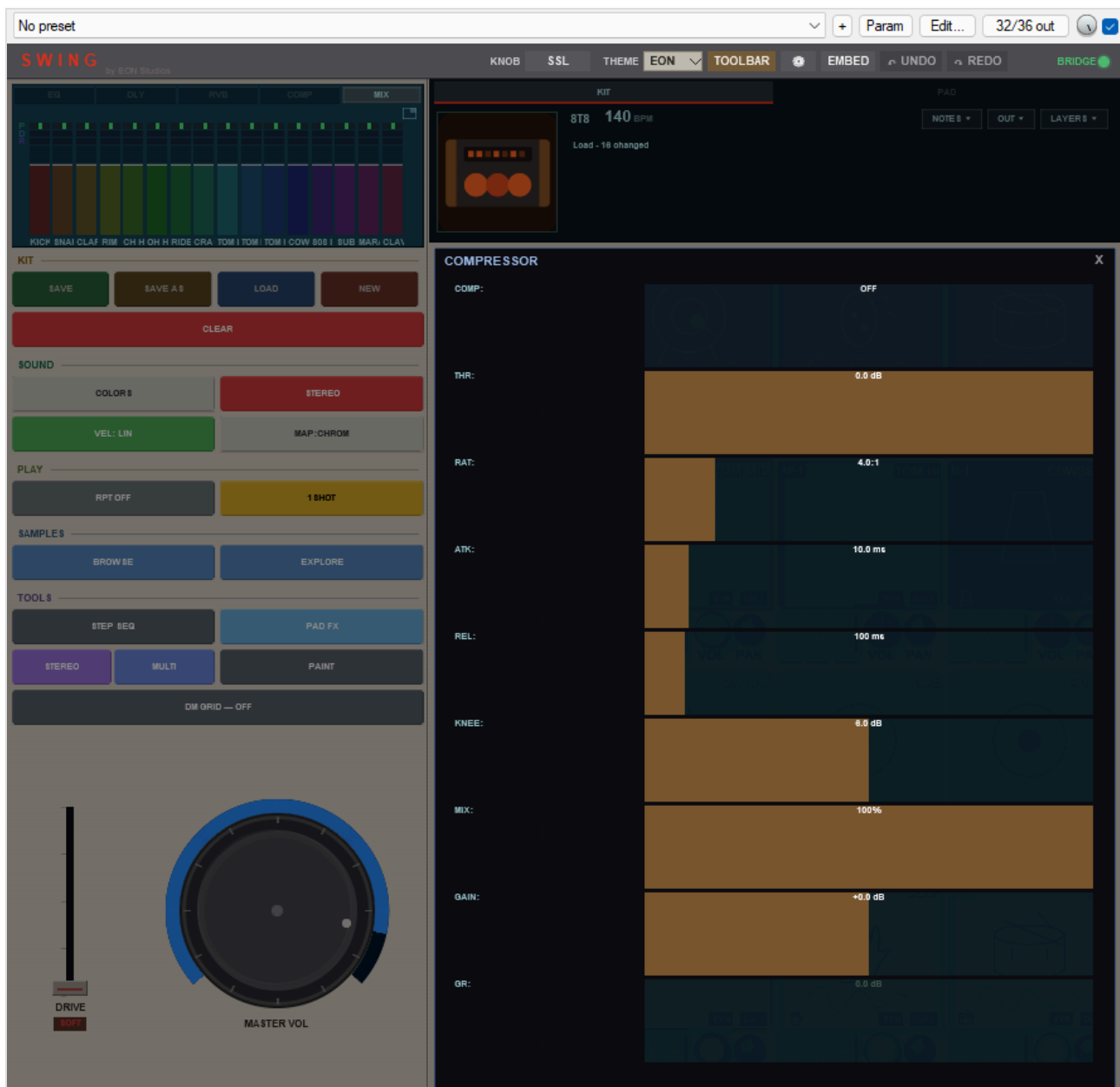
Delay. Tempo-synced, with feedback and tone controls.



Reverb. Size, damping, width and pre-delay.



Compressor. Threshold, ratio, attack, release, knee — glue and punch.



Limiters / master volume. Keep peaks under control, and level-match when comparing presets.

11. Swing Kit Bridge

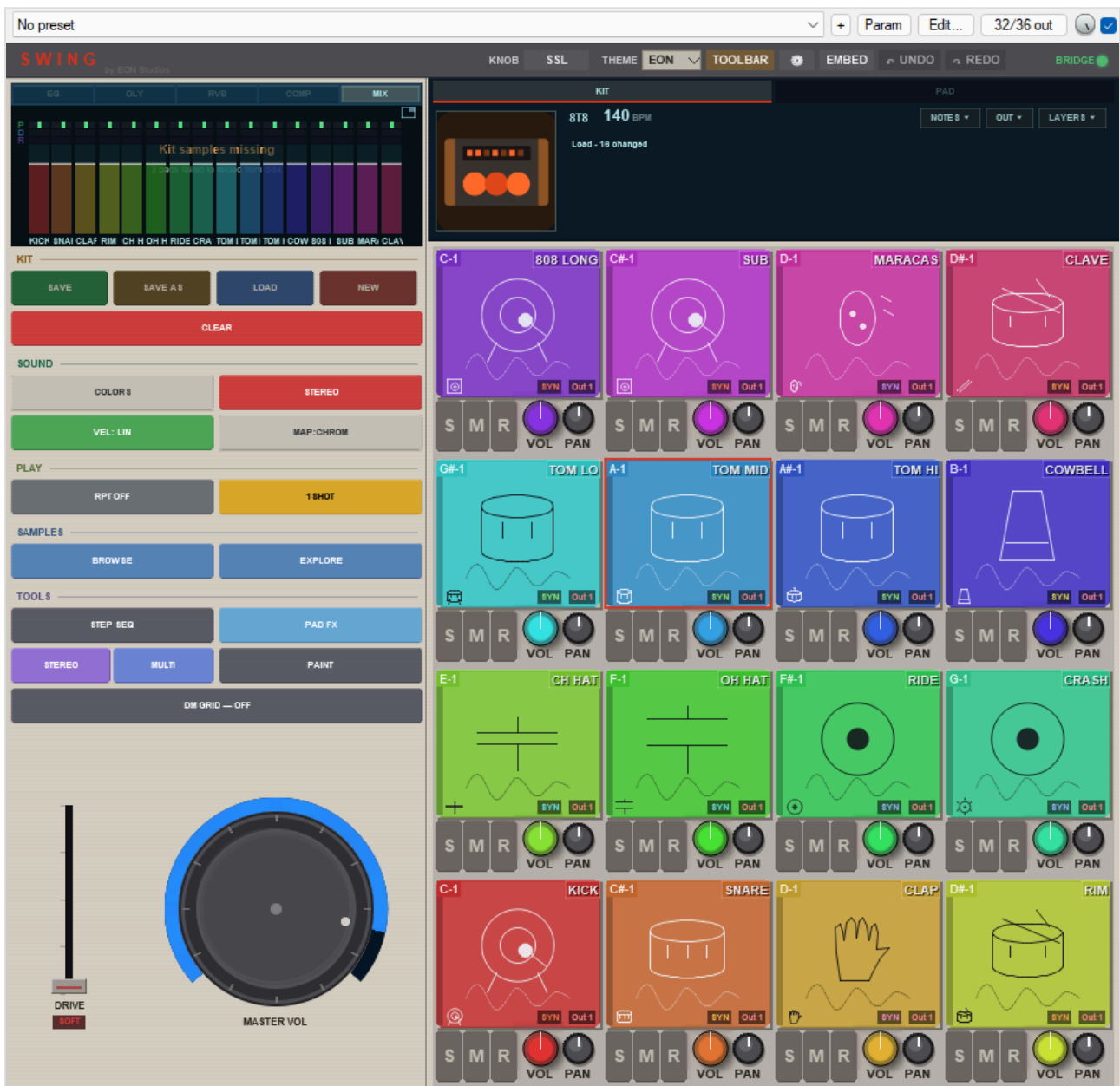
The Kit Bridge is a background Lua service that enables kit save/load and other operations JSFX alone cannot do safely. It registers itself on install; you can re-run it any time from the Action List.

Save/Load kits. Writes and reads `.swing` files, including pad names, colours, output routing, ADSR, FX settings, synth voices and cover artwork.

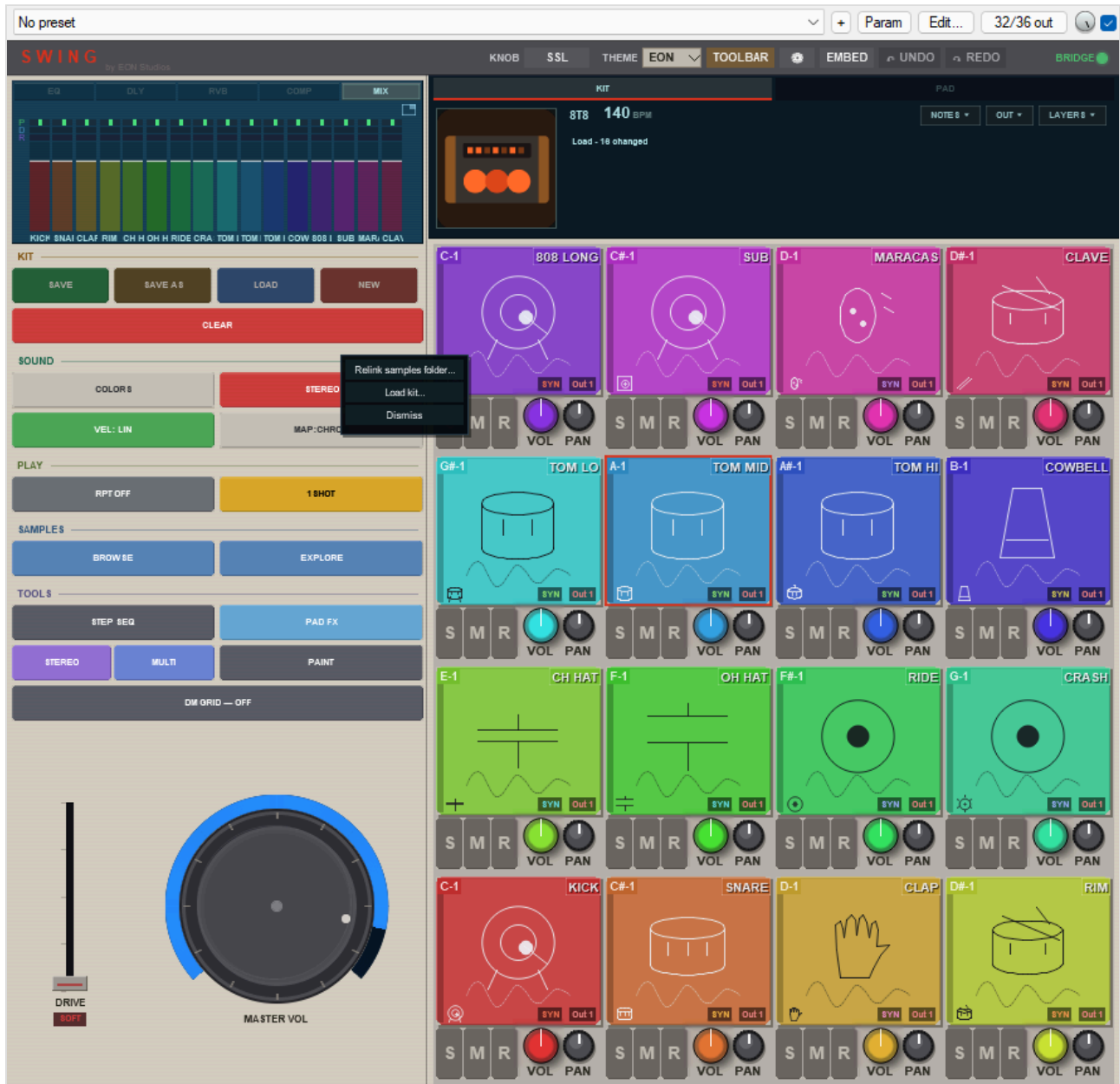
Batch import. Load a whole folder of samples and auto-distribute to pads.

Chop to pads. Slice an arrangement selection straight onto pads, from the action list or the LOAD right-click menu.

Relinking. *New in v3.* When samples go missing — you moved or renamed a folder — Swing says so instead of leaving silent pads.



Clicking the banner offers to relink the folder, load a different kit, or dismiss.



If a whole project moves, Swing re-points automatically.

Embed audio: Auto / Always / Never. *New in v3.* Decides whether pad audio is written inside the project file. *Auto* keeps projects small; *Always* makes a project self-contained and portable, at the cost of size.

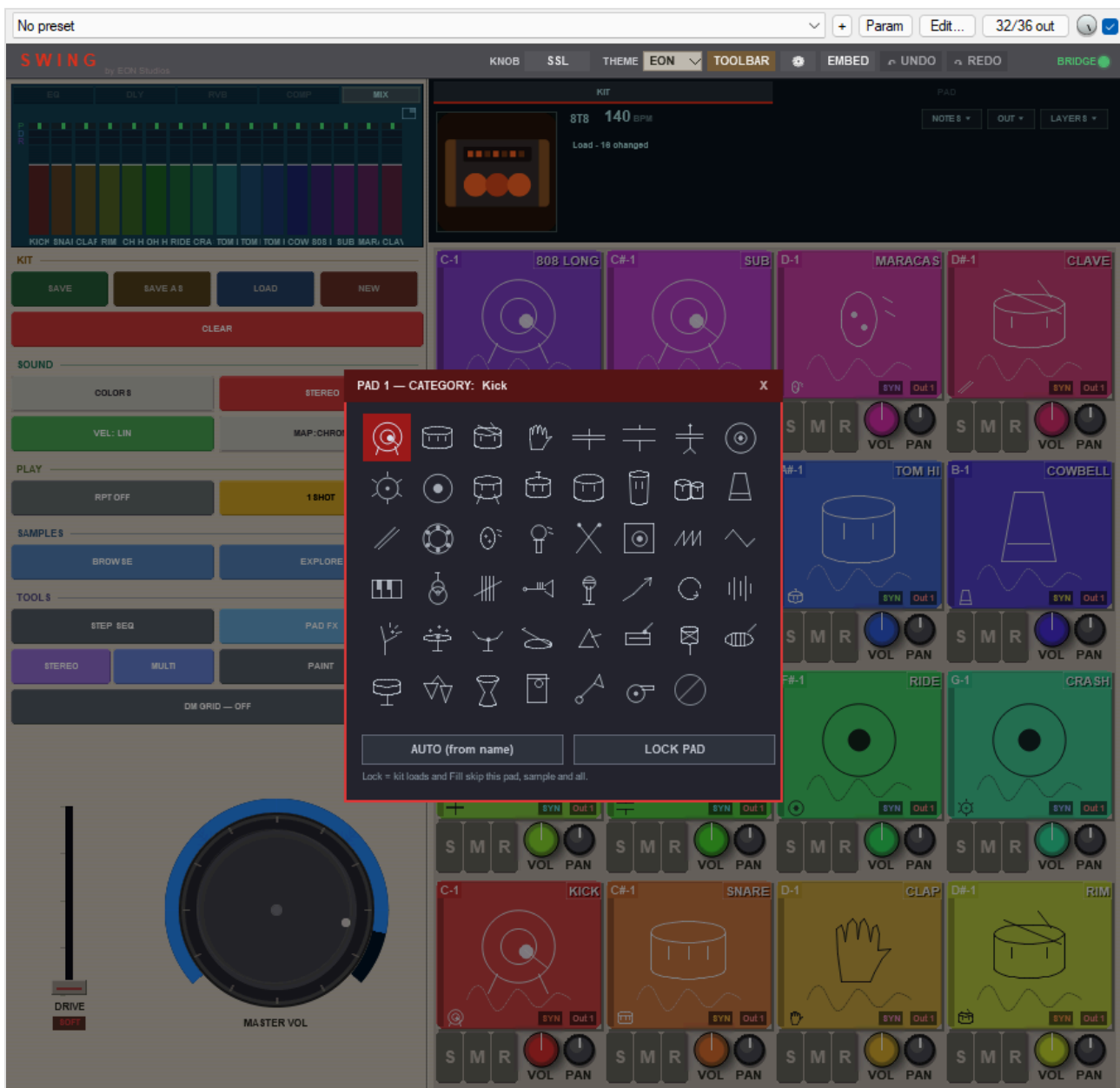
Housekeeping. Pad audio that was never saved into a kit or project, and has gone untouched for 30 days, is deleted automatically so the scratch store does not grow without limit. This only ever touches Swing's own unsaved scratch audio — kits, projects and your own sample files are never involved.

12. Kit Categories & Artwork

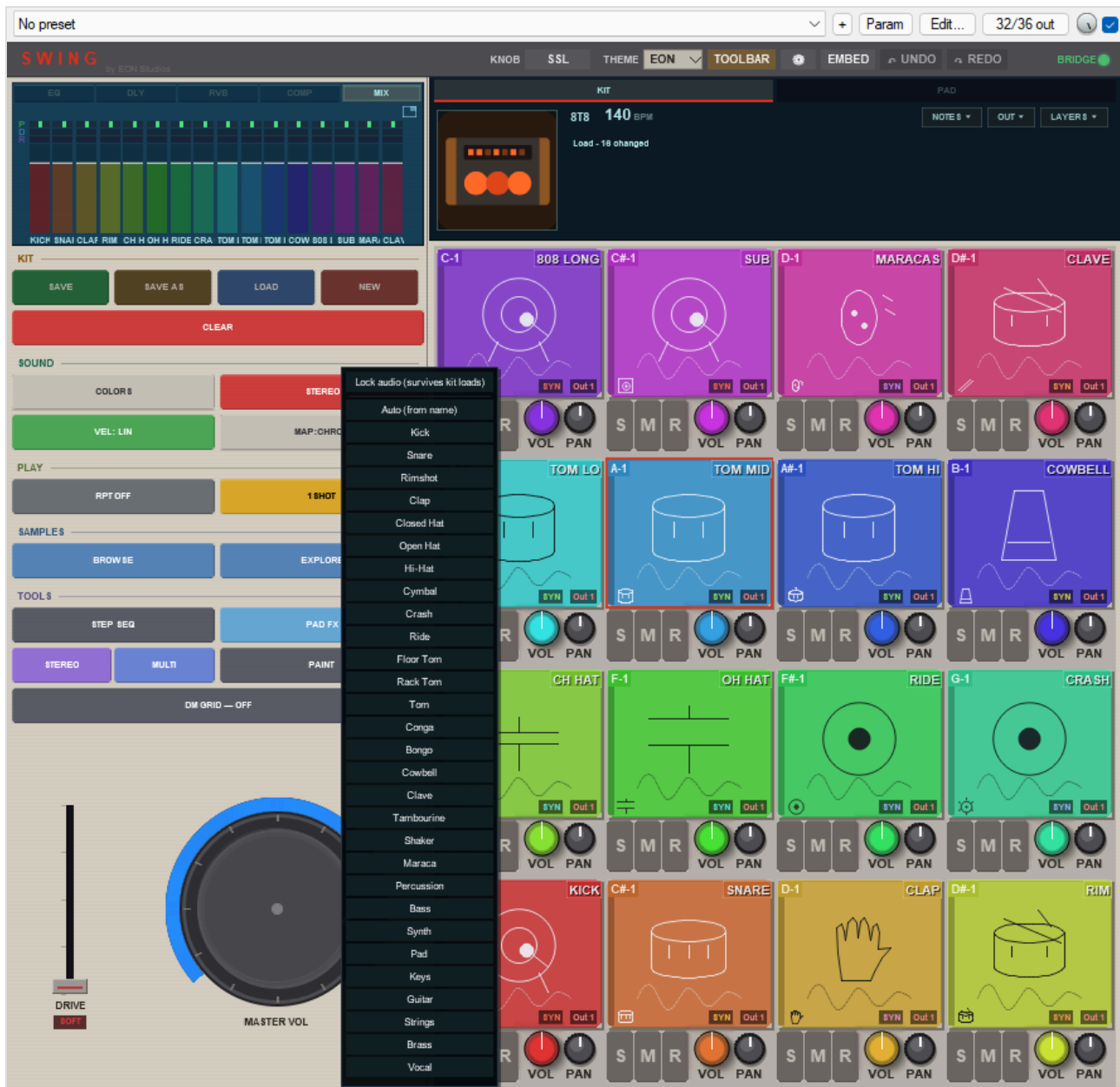
New in v3.

Categories, LOCK and ADAPT

Each pad carries a category glyph in place of its number, picked from a grid of 46.



The same vocabulary is available as a list, with LOCK at the top:



LOCK keeps a pad exactly as it is through kit loads — locked pads are untouched by anything a new kit brings.

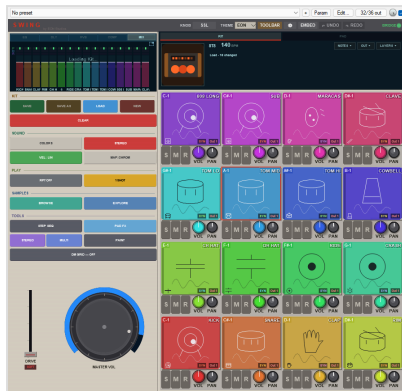
ADAPT re-aims your existing patterns at the new kit rather than leaving them pointing at the old one. A change map shows what moved after a load.

Kit artwork

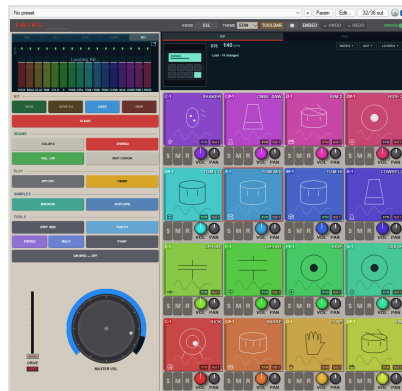
Kits carry a cover image inside the `.swing` file, so it travels with the kit wherever the kit goes. Drag a **PNG** or **JPEG** onto the KIT tab's cover tile to attach one. The new cover appears immediately, but it is only written into the `.swing` file when you next save the kit.

All thirteen factory kits ship with machine-portrait covers.

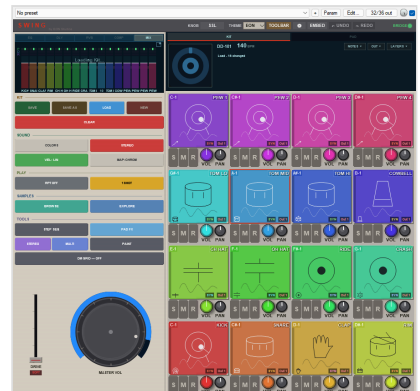
8T8



DD-101



SPACE



13. Swing Browser

The Swing Browser is an ImGui-based sample browser with folder navigation and audio preview.



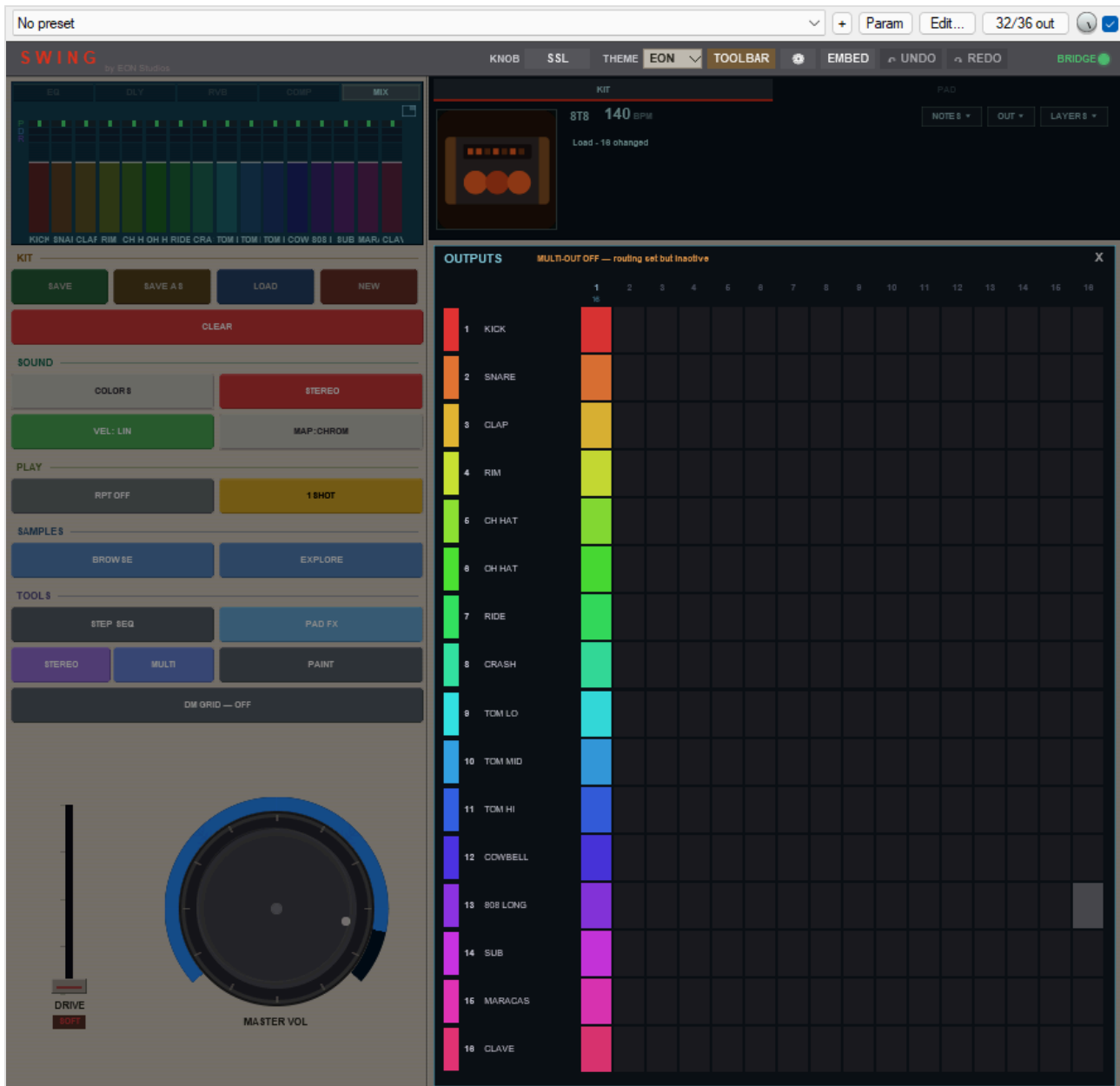
Preview. Audition samples before loading. Waveform, BPM and key detection help organise large libraries.

Drag & drop. Drag files directly onto pads — including onto a Swing embedded in the track panel or mixer. Multi-file drags fill consecutive pads.

Targeting. In projects with multiple Swings, the browser targets a specific instance, so a dropped sample lands where you meant it to.

14. Multi-Out Routing

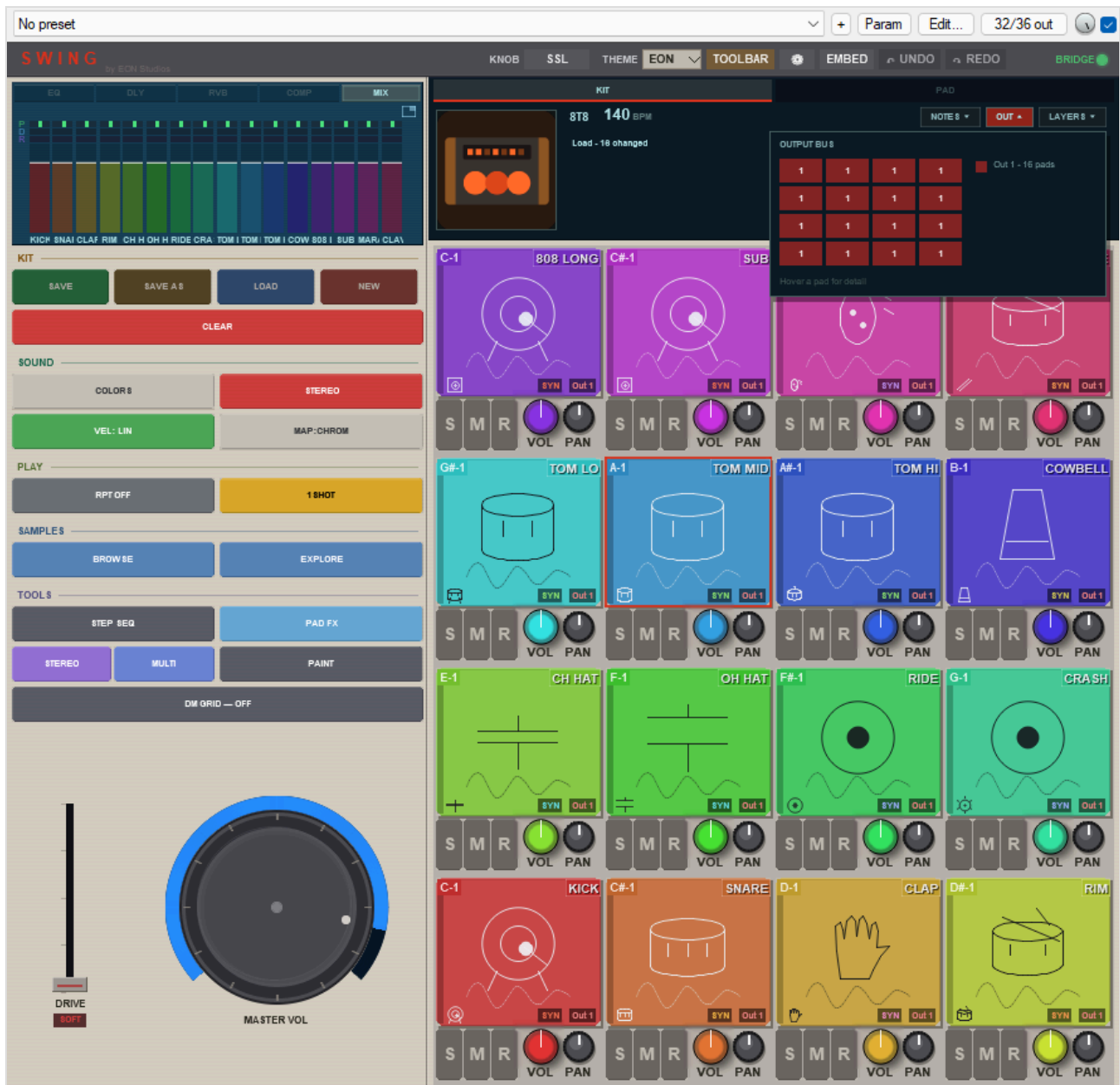
Swing supports 16 stereo output buses — one per pad if you like.



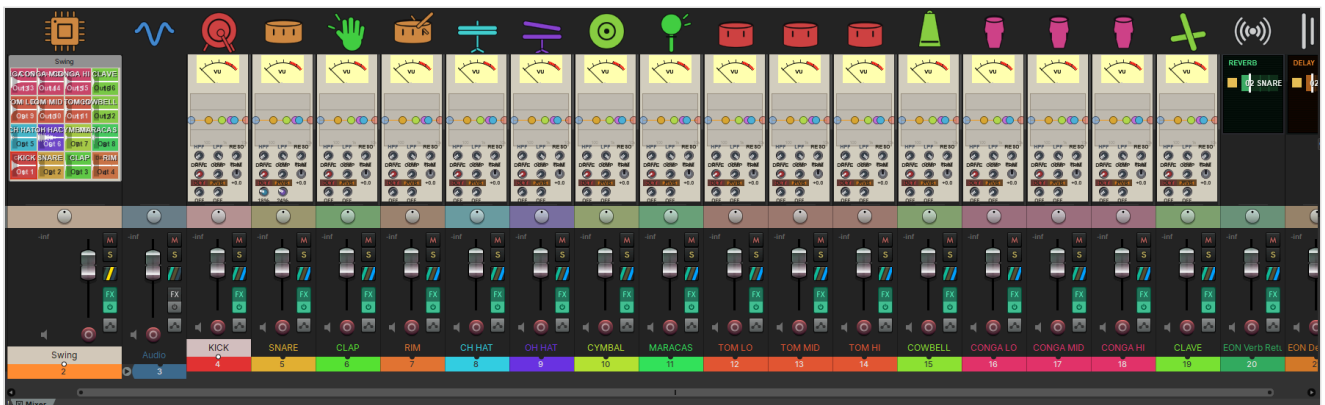
Build the rig. Right-click the **STEREO / MULTI** buttons in the tools row and choose **Build Multi-Out (Audio)**. That creates the matching REAPER tracks so each pad appears on its own mixer channel.

The same menu offers **Build Stereo Grid**, **Build Drum Matrix (MIDI lanes)** and **Build Both (Audio + MIDI)**.

The KIT tab's **OUT** panel shows all sixteen buses at a glance:



Built, it looks like this — a channel per pad, each carrying its own Drum Strip, with the pad's colour and category icon carried through to the track:



The two rightmost channels are the reverb and delay returns.

Mixing. Once built, each pad has its own channel in REAPER's mixer — add EQ, compression and sends per drum there while keeping Swing's pad engine simple.

FX returns. *New in v3.* Opt-in delay and reverb return outputs for multi-out rigs, with per-pad send knobs on the Drum Strip.

15. Step Sequencer, Drum Matrix & Drum Strip

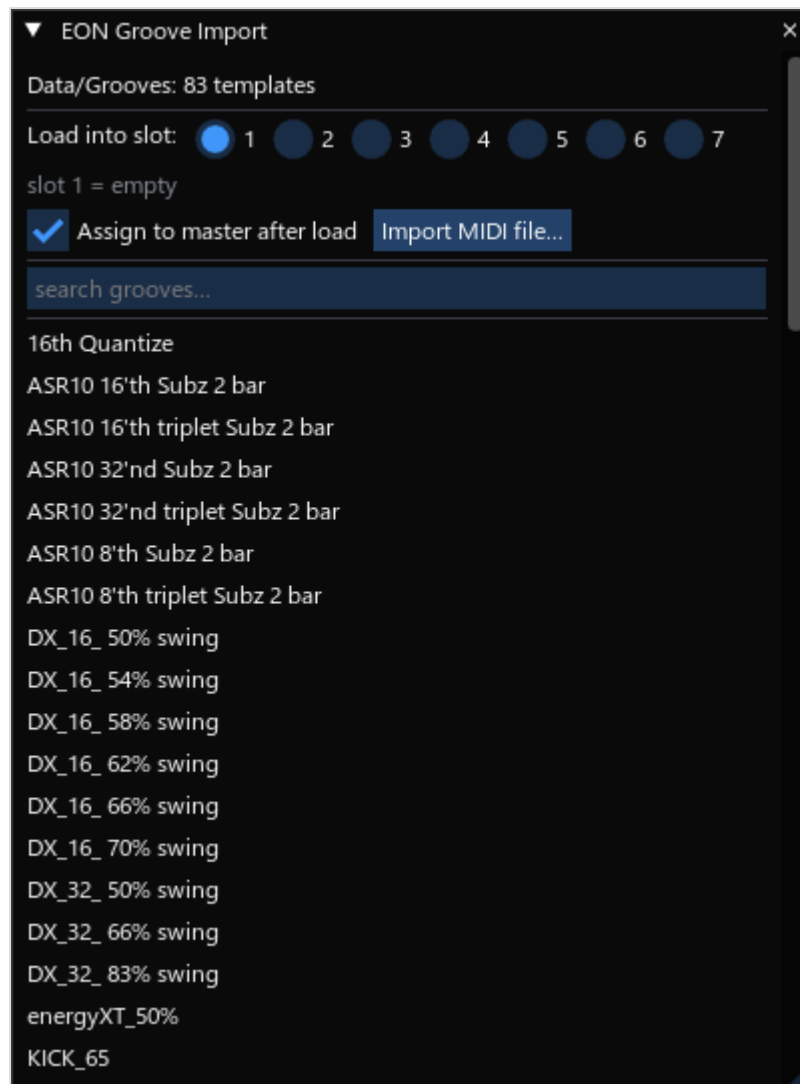
All three are new in v3. Version 2.1 installed Swing and nothing else.

Step Sequencer

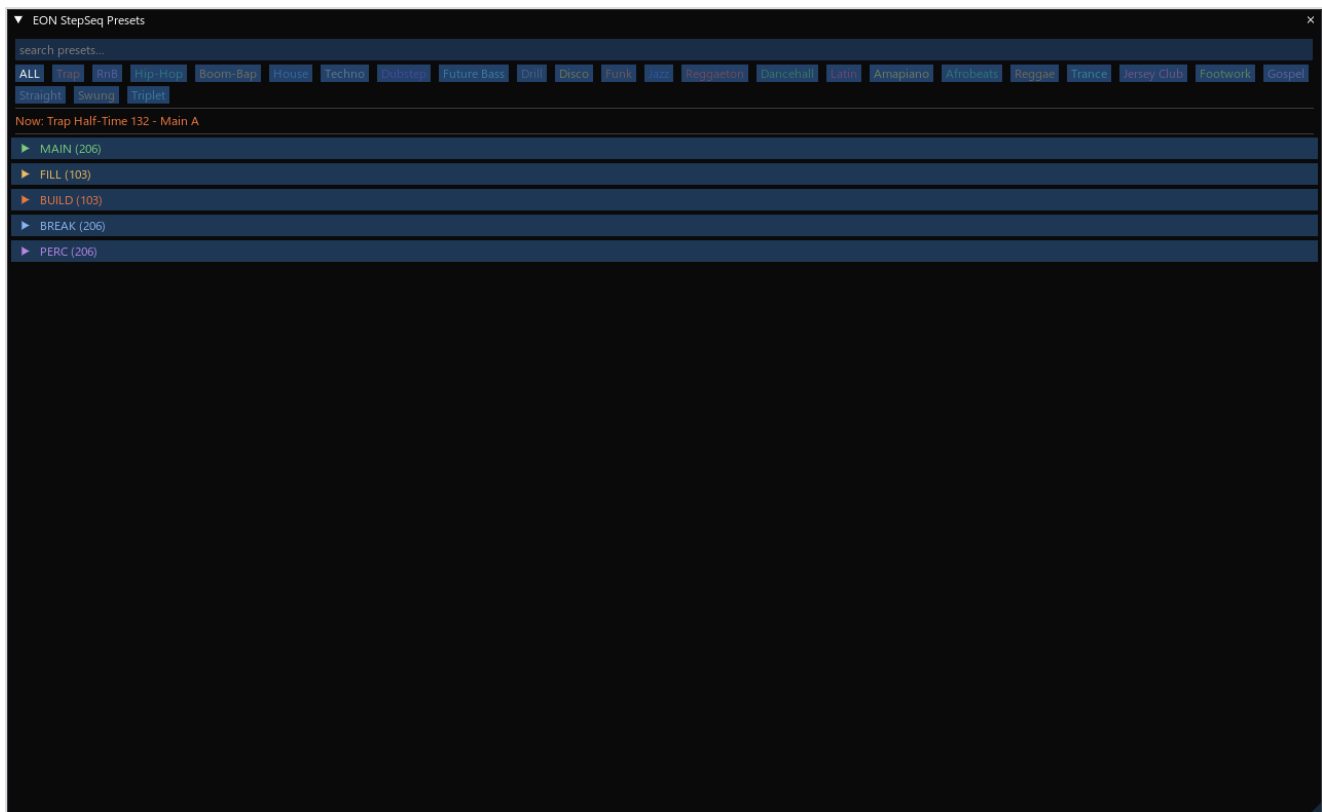


Pattern sequencing with per-lane volume, pan and groove.

Groove system. A swing and groove engine with `.rgt` import *and* export, plus groove import straight from a MIDI file.



Preset browser, organised by role — main, fill, build, break, perc — with genre chips and live counts.



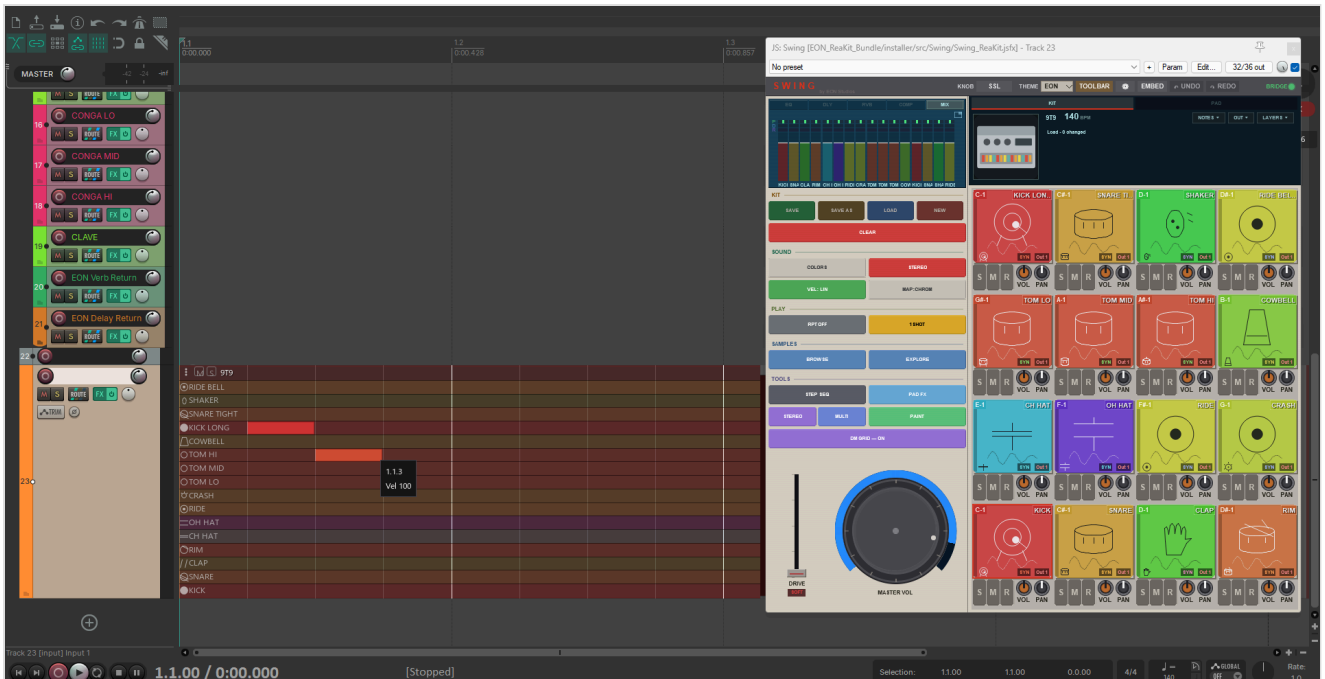
Song mode. Named arrangements that switch at the play-next boundary.

Drum Matrix

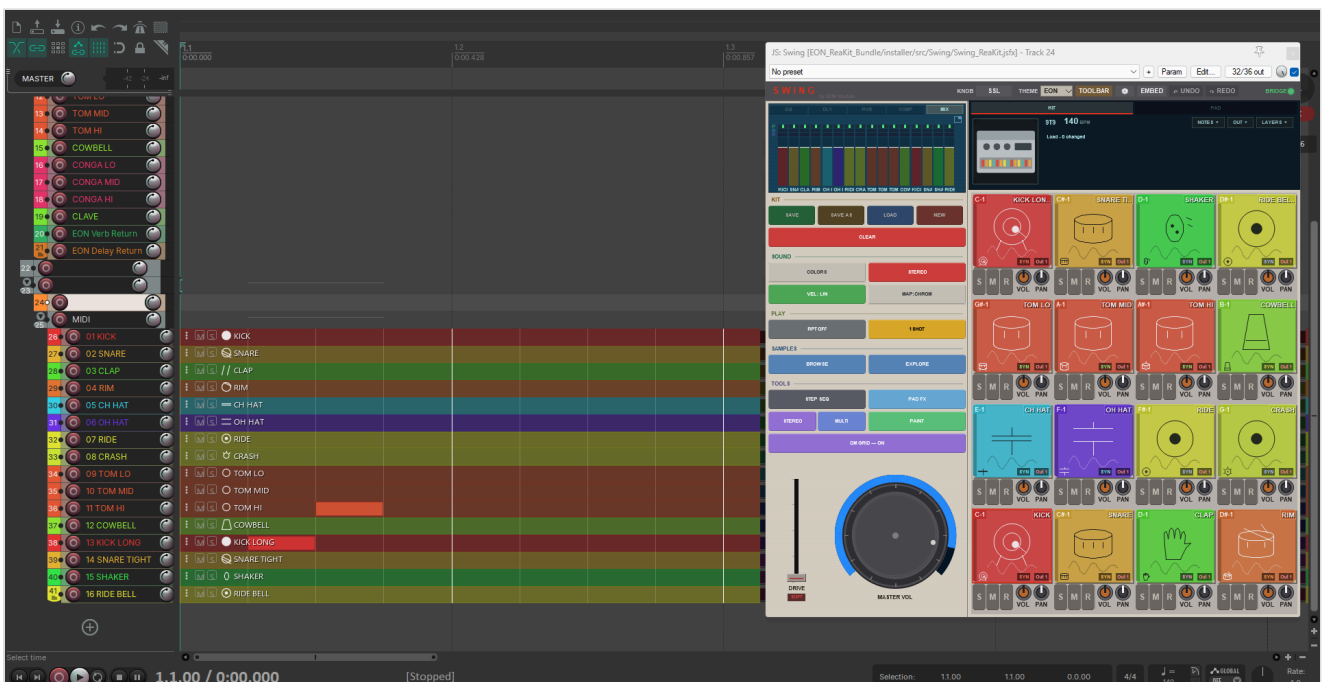
Your patterns as a grid in the arrange window, working on real REAPER MIDI items rather than a private format. Lanes carry your pad names, colours and icons straight through from Swing. Edit either end — sequencer or matrix — and the other follows, ratchets included. Role-based remapping means a pattern aimed at one kit lands sensibly on another.

Two ways to lay it out

Stereo grid keeps everything on the Swing track: one MIDI item, with a lane per pad inside it. Nothing else is added to your project. Build it from the tools row with **STEREO**, or right-click → **Build Stereo Grid (this track)**.



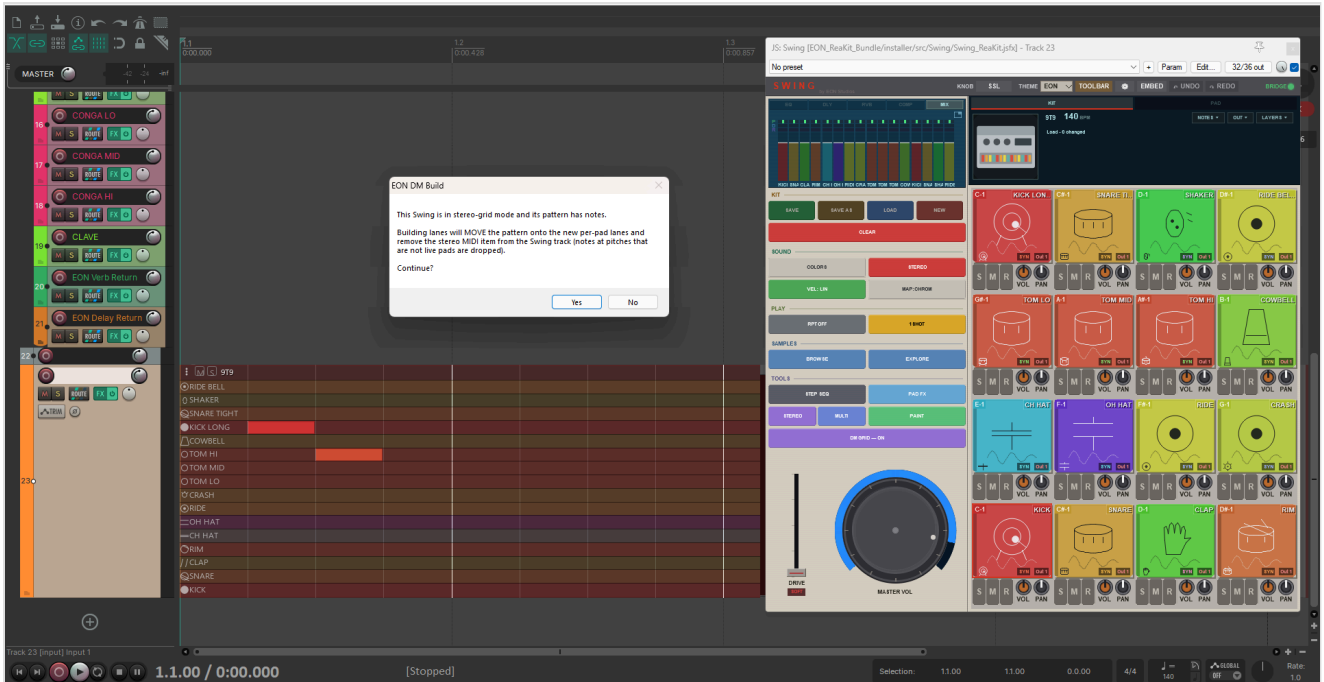
MIDI lanes gives each pad its own REAPER track — sixteen of them, named and coloured from the kit. Build it with **MULTI**, or right-click → **Build Drum Matrix (MIDI lanes)**.



The right-click menu on those buttons also offers **Build Multi-Out (Audio)** and **Build Both (Audio + MIDI)**.

Switching from stereo grid to lanes

If the Swing is already in stereo-grid mode *and* its pattern has notes, Swing stops and asks before going any further, because the change is not additive:



Building lanes will **move** the pattern onto the new per-pad lanes and remove

the stereo MIDI item from the Swing track. **Notes at pitches that are not live

pads are dropped.**

So a pattern written against a kit with empty pads, or against a different kit, can lose notes in the move. Answer No and nothing is touched.

When the build finishes it tells you how many lanes are ready and asks you to **load or restart** `eon_drum_matrix.lua` to display them — the lanes exist in the project either way, but the Drum Matrix window has to be re-opened to pick them up.

Drum Strip



The per-pad chain as its own plugin, and a full embedded track and mixer view. SSL-style knobs, EQ graph, gain-reduction meter, HPF/LPF nodes and VU-face selectors. One strip can broadcast its layout to the others, so a whole kit's strips match in one move.

FX Return View

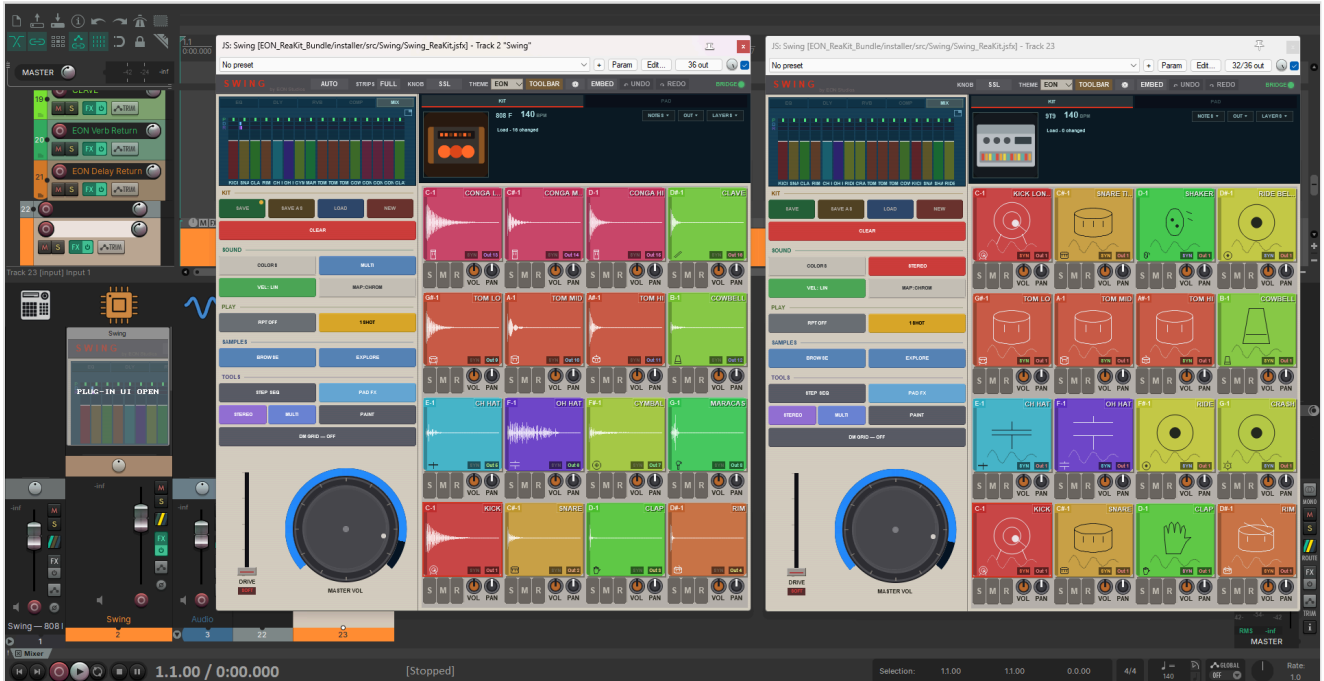
Shows which pads are feeding the delay and reverb returns, and how hard.

16. Multi-Instance & Undo

New in v3.

Several Swings in one project

Every instance is its own instrument. Each keeps its own pitch state, mute and solo, colours and note names, and syncs two-way with the Drum Matrix independently. Nothing leaks between them.



Here the same project runs Swing on track 2 with the Fischer 808 kit and on track 23 with the 9T9 synth kit — different pads, different notes, different output routing, no interference.

The Sample Browser targets one instance at a time, so a dropped sample lands where you meant it to.

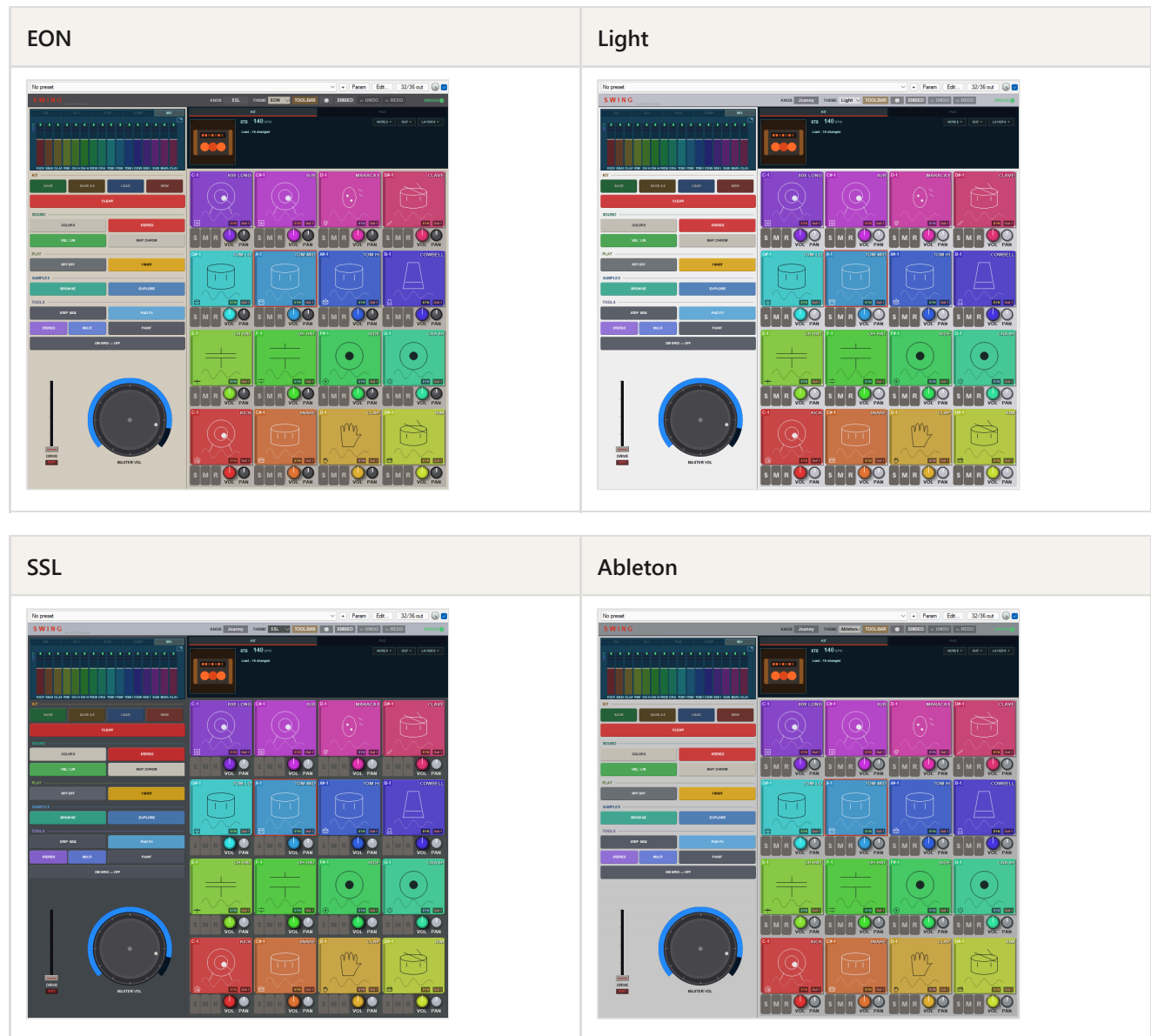
Undo

Swing has its own undo engine — parameter, structural and full-kit — pulled out of REAPER's global undo. Ctrl+Z can no longer wipe a kit.

17. Themes, Knobs & Controllers

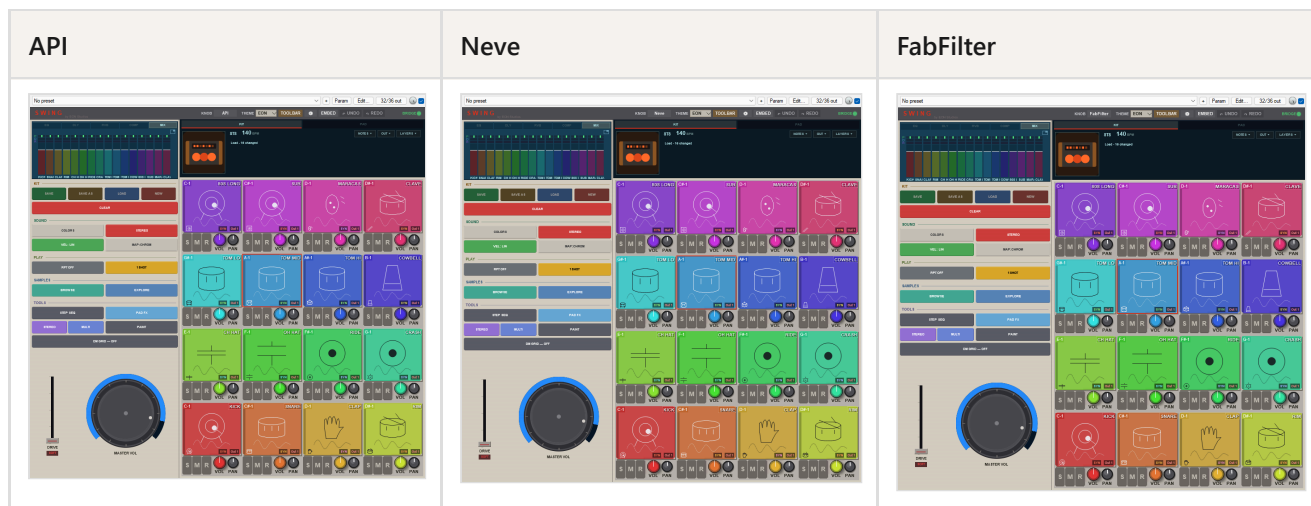
Themes

Fourteen console themes, each shipping its own knob style.



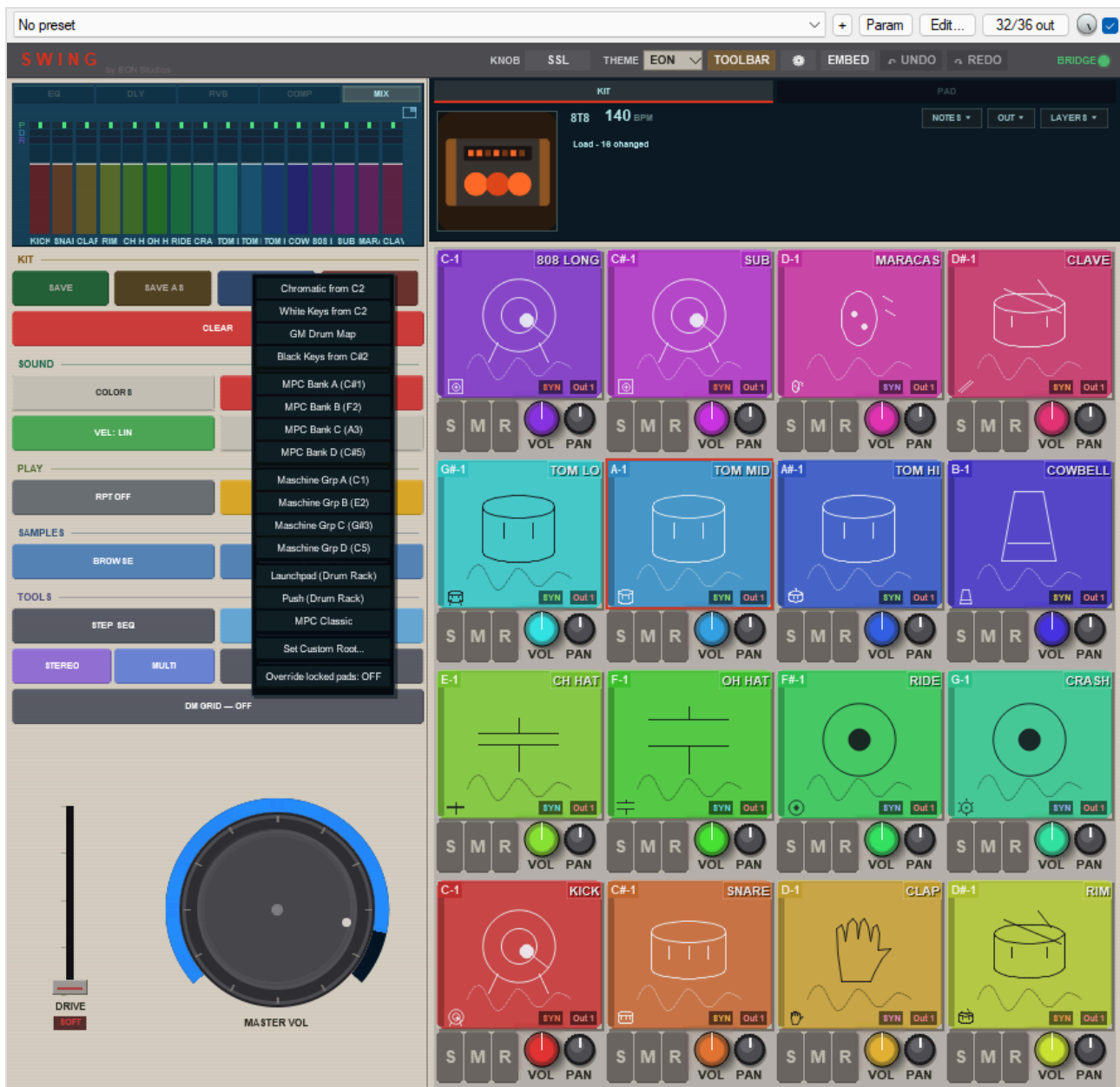
Knob styles

Knob style is separately selectable from the theme — eighteen of them.



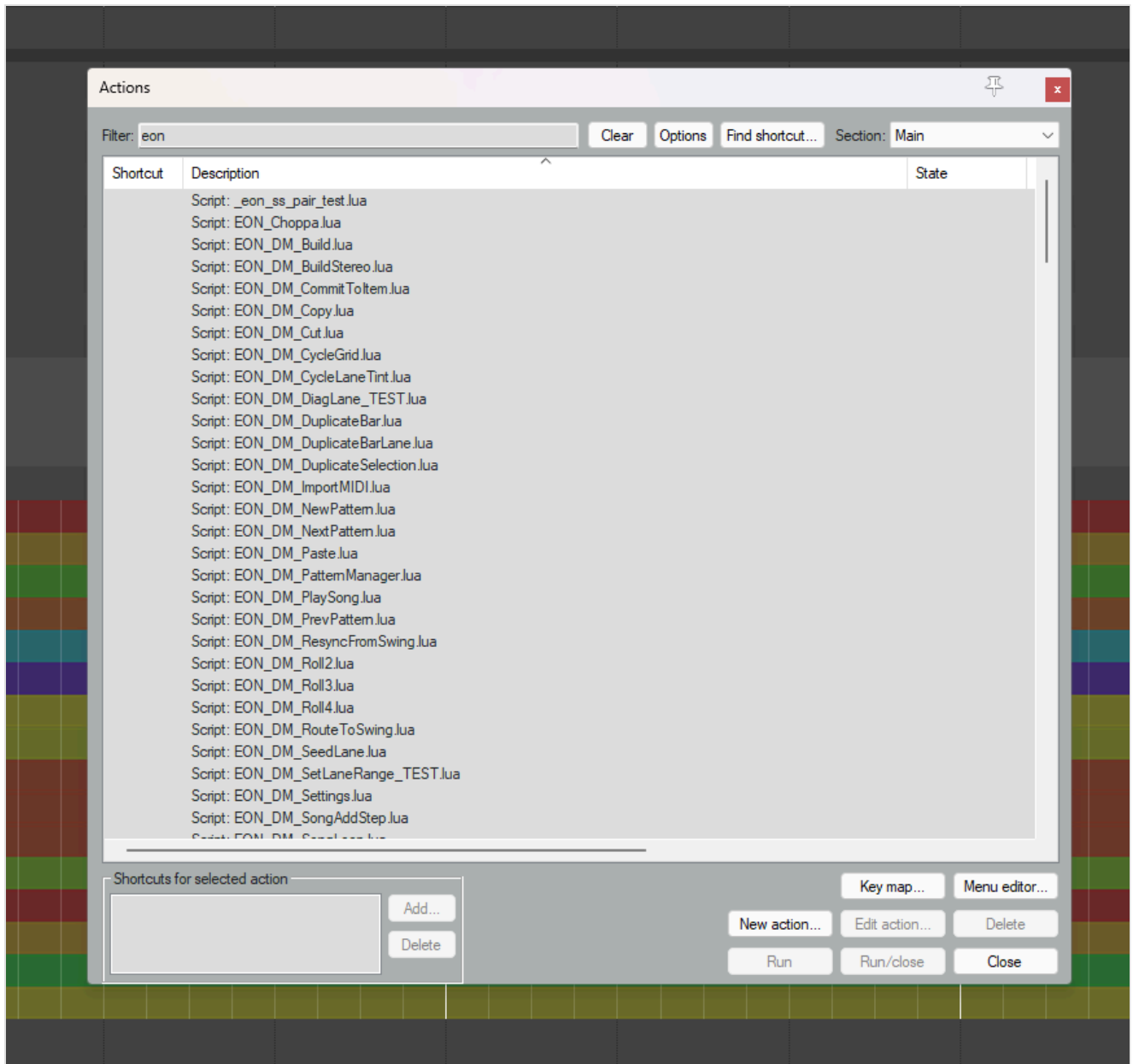
Per-pad knob colour. Right-click a pad's VOL or PAN knob — left half is VOL, right half is PAN — for that knob's own style and colour. **Follow Pad** tints the knob with the pad's hue. The two knobs are independent and the choice persists.

Controller presets



Sixteen presets covering MPC, Maschine, Launchpad and Push, plus a custom root note and an override-locked-pads toggle. A MIDI-learn mode steps through the pads so you can map a controller in one pass.

Actions and the toolbar



The whole catalogue is keybindable and toolbar-ready, and registers itself on install. New in v3: **New Song** (lays out tempo, four named regions, a Swing track, a MIDI item per region and the drum lanes in one action), **Install / Refresh Toolbar**, and ten view actions.

18. Recipes

Starting points, not rules.

Tight kick. Offset forward to remove silence at the front; HPF 25–35 Hz; a touch of Tape drive; keep decay short if the sample is long.

Punchy snare. Mid EQ +2–4 dB around 1–3 kHz; Soft or Tube drive in small amounts; short room reverb send.

Lo-fi hats. Bit depth 8–12 bits; gentle LPF to tame aliasing; small reverb send.

808 sub. Minimal drive; low shelf EQ +1–2 dB; long release; route to its own output to manage the low end.

Wide clap. Slight pan spread across layers in Sum mode; reverb width up; pre-delay for separation.

A synth kick from scratch. Switch the pad to synth and pick the kick family. Start with tune and decay — those two get you most of the way. Add drive last; it is easy to overdo when there is no sample noise to hide behind.

Hybrid pad. Layer a synth voice under a sample for weight the sample lacks. Set the synth low and let the sample keep the transient.

A playable 808. Give the pad a key range, leave it in Repitch, and play the line — one note at a time, since ranged pads are monophonic.

Hat pair. Put the open and closed hats in the same choke group. That is the whole trick.

Parallel crush. Send a pad into a return and crush the return rather than the pad, so the clean hit stays underneath.

19. Troubleshooting

Plugin not found. Restart REAPER. Verify files exist under `Effects/EON/Swing` and re-scan if needed.

Browser won't open. Requires RealmGui and SWS — install via ReaPack if missing.

Drum Matrix won't open. Requires js_ReaScriptAPI, via ReaPack.

Kit Bridge doesn't start. It registers as a startup action after install. You can also run `Swing_Kit_Bridge` manually from the Action List; it re-registers itself on first run.

Stretch sounds like Repitch. The EON engine is not installed. Tuned and Stretch fall back to Repitch without it, and Swing shows a notice saying so.

Old kits sound different. Swing loads 2.1-era kits with neutral defaults for anything that did not exist then — filters wide open, EQ flat, saturation off, no sends.

A pad plays only one note of a chord. Expected. Ranged pads are monophonic.

Pads came back blank after reopening a project. Fixed in v3. If you see it on a 3.0 install, check that the Kit Bridge is running.